

ALIBABA CLOUD INFRASTRUCTURE GUIDE FOR GOOGLE CLOUD PLATFORM (GCP) PROFESSIONALS

CLOUD MIGRATION
WHITEPAPER SERIES

 Alibaba Cloud

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TARGET AUDIENCE

This guide is intended to provide Google Cloud (GCP) professionals with methodologies and best practices to understand Alibaba Cloud infrastructure, whether you are planning a multi-cloud solution with GCP and Alibaba Cloud or migrating fully to Alibaba Cloud.





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1. WHY MAKE THE MOVE?

Here are some key considerations that may drive customers to make the move from GCP to Alibaba Cloud:

GO CHINA & GO ASIA

Alibaba Cloud was born and raised in China, which strengthens the understanding of both the complexities and opportunities of the Chinese market, allowing us to offer customers outstanding “Go China” solutions. Especially considering the fact that GCP currently has no region in China yet. Please read more about [China Gateway](#).

As a native Asian company and Asia's No. 1 Infrastructure as a Service (IaaS) service provider recognized by Gartner, Alibaba Cloud can accelerate your success in Asia today. Alibaba Cloud has the most data centers (20 regions, 71 zones) in the Asia Pacific area than any other public cloud provider. Please read more about [Asia Accelerator](#).

COST EFFICIENCY

Alibaba Cloud offers flexible, easy-to-use, and cost-effective resource packages to further save your cost, such as monthly subscriptions for many Alibaba Cloud products, Storage Capacity Units (SCU) for storage, Data Transfer Plans for network traffic, and 95 percentile billing for Cloud Enterprise Network (CEN), etc. Please read more about [Alibaba Cloud pricing](#).

COMPREHENSIVE PORTFOLIO

Alibaba Cloud offers comprehensive and high-performance cloud products, including large-scale computing, storage resources, and big data processing capabilities for users around the world. Alibaba Cloud has more instance types of virtual machines, cross-region network products, types of databases, and middleware services such as MQ (message queue) for Apache Kafka, RabbitMQ, and RocketMQ.



MULTI-CLOUD / HYBRID CLOUD

Alibaba Cloud is the pilot of hybrid clouds and has supported many customer successes across many industries. Inspired by the Alibaba Cloud public cloud, Alibaba Cloud Hybrid Cloud is equipped with over 50 full-stack product portfolios and solutions, enabling you to practice hybrid cloud in various business scenarios. Please visit the Alibaba Cloud website to learn more about Alibaba Cloud [Hybrid Cloud](#) solutions.

SECURITY AND COMPLIANCE

The security process of Alibaba Cloud has been recognized by authorities both inside and outside mainland China. By leveraging its years of expertise in defending against Internet security threats, Alibaba Cloud provides security protection for the cloud platform and integrates multiple compliance standards into the internal control and product design of the cloud platform. Please read more about Alibaba Cloud's [security & privacy compliance](#).

User data security on the cloud is the lifeline of users and one of the most important representational attributes of the overall cloud security capability. In July 2015, Alibaba Cloud initiated the first Data Protection Proposal among cloud service providers in China. This public proposal appeals that the ownership of data of all developers, companies, governments, and social institutions on the cloud computing platforms belongs only to the users. Cloud computing platforms cannot use the data for other purposes. Cloud service providers have a responsibility and obligation to help users ensure the confidentiality, integrity, and availability of their data. Please read more about Alibaba Cloud's [data security](#).

INSTANT SUPPORT

Alibaba Cloud offers highly flexible support services tailored to meet your exact needs. To better support enterprise customers, a dedicated technical account manager (TAM) would be assigned along with the creation of a dedicated instant messaging DingTalk group. Please read more about Alibaba Cloud's [after-sales support](#).





2. SERVICES COMPARISON BETWEEN GCP AND ALIBABA CLOUD

As of January 2022, Alibaba Cloud has 80 zones strategically located across 25 regions in the world, providing reliable computing support for billions of customers. Please visit the Alibaba Cloud website to check our global [Data Centers](#).

Area	Country	City	Region Id	Number of Zones
Asia Pacific (20)	China	Beijing	cn-beijing	12
		Chengdu	cn-chengdu	2
		Guangzhou	cn-guangzhou	2
		Hangzhou	cn-hangzhou	8
		Heyuan	cn-heyuan	2
		Huhhot	cn-huhehaote	2
		Nanjing	cn-nanjing	1
		Qingdao	cn-qingdao	2
		Shanghai	cn-shanghai	9
		Shenzhen	cn-shenzhen	6
		Ulanqab	cn-wulanchabu	3
		Zhangjiakou	cn-zhangjiakou	3
		Hong Kong	cn-hongkong	3
	Japan	Tokyo	ap-northeast-1	2
	India	Mumbai	ap-south-1	2
	Singapore	Singapore	ap-southeast-1	3
	Australia	Sydney	ap-southeast-2	2
	Malaysia	Kuala Lumpur	ap-southeast-3	2
	Indonesia	Jakarta	ap-southeast-5	3
	Philippines	Manila	ap-southeast-6	1
Europe & Americas (4)	Germany	Frankfurt	eu-central-1	2
	UK	London	eu-west-1	2
	US	Silicon Valley	us-west-1	2
		Virginia	us-east-1	2
Middle East (1)	UAE	Dubai	me-east-1	1



Alibaba Cloud offers comprehensive products and services. For most of the GCP products available, customers can find the equivalent on Alibaba Cloud. Below is a services comparison between GCP and Alibaba Cloud IaaS and database products.

2.1 ELASTIC COMPUTING

2.1.1 CLOUD SERVER

GCP Product	Alibaba Cloud Product	Description
Compute Engine	Elastic Compute Service (ECS)	Elastic and secure virtual cloud servers to cater to all your cloud hosting needs.
Elastic Bare Metal	ECS Bare Metal Instance (EBM)	An elastic and horizontally scalable high-performance computing service providing the same computing performance as traditional physical servers including physical isolation.
Cloud GPUs	Elastic GPU Service (EGS)	Powerful parallel computing capabilities based on GPU (Graphics Processing Units) technology.
Sole-tenant nodes	Dedicated Host (DDH)	A dedicated resource pool that offers flexible and compliant deployment options to grant enterprise users with more control over their cloud infrastructure.
Virtual Desktops (3rd Party)	Elastic Desktop Service (EDS)	A convenient and secure cloud-based Desktop as a Service (DaaS) solution.

2.1.2 ELASTIC ORCHESTRATION

GCP Product	Alibaba Cloud Product	Description
Compute Engine Autoscaler	Auto Scaling	Automatically adjusts computing resources based on your business cycle.
Cloud Deployment Manager	Resource Orchestration Service (ROS)	A fully managed service for cloud computing resource orchestration and automated deployment.
VM Manager	Operation Orchestration Service (OOS)	Automated operations and maintenance (O&M) service.

2.1.3 HIGH-PERFORMANCE COMPUTING

GCP Product	Alibaba Cloud Product	Description
High performance computing	Elastic High Performance Computing (E-HPC)	A HPCaaS (High-Performance Computing as a Service) cloud platform providing an all-in-one high-performance public computing service.





2.2 NETWORKING

2.2.1 CLOUD NETWORK

GCP Product	Alibaba Cloud Product	Description
Cloud Virtual Network	Virtual Private Cloud (VPC)	A virtual private cloud service that provides an isolated cloud network to operate resources in a secure environment.
Cloud Load Balancing	Server Load Balancer (SLB)	Server Load Balancer (SLB) distributes network traffic across groups of backend servers to improve the service capability and application availability. It provides functions as a reverse proxy at Layer 7 (ALB – Application Load Balancer) and load balancing services at Layer 4 (CLB – Classic Load Balancer).
Cloud NAT	NAT Gateway	A Network Address Translation (NAT) gateway enables multiple instances within a VPC to communicate with the Internet. Custom Source Network Address Translation (SNAT) and Destination Network Address Translation (DNAT) rules can be created to help you use network resources in a flexible manner.
Private Service Connect	PrivateLink	Connect your VPCs to services in other VPCs through secure, reliable, and private connections. PrivateLink simplifies network architecture and reduces security risks because network traffic does not traverse the Internet.

2.2.2 CROSS-REGION NETWORK

GCP Product	Alibaba Cloud Product	Description
Network Connectivity Center	Cloud Enterprise Network (CEN)	A global network for rapidly building a distributed business system and hybrid cloud to help users create a network with enterprise-level scalability, and with the communication capabilities of a cloud network.
Realized by Anycast+CLB. No independent product	Global Accelerator (GA)	Provides a network acceleration service for your Internet-facing application globally with guaranteed bandwidth and high reliability.



2.2.3 HYBRID CLOUD NETWORK

GCP Product	Alibaba Cloud Product	Description
Cloud InterConnect	Express Connect	An easy-to-use network service that enables high-bandwidth, reliable, secure, and private connections between different environments.
Cloud VPN	VPN Gateway	Virtual Private Network (VPN) Gateway is an Internet-based service that establishes a connection between a VPC and your on-premise data center.

2.3 STORAGE

2.3.1 DATA TRANSPORT

GCP Product	Alibaba Cloud Product	Description
Transfer Appliance	Data Transport	A secure solution to migrate TB-level (Terabyte-level) or PB-level (Petabyte-level) data to Alibaba Cloud.

2.3.2 ESSENTIAL STORAGE SERVICE

GCP Product	Alibaba Cloud Product	Description
Persistent Disk	Elastic Block Storage (EBS)	Block-level data storage attached to ECS instances to achieve high performance, low latency, and high reliability.
Cloud Storage	Object Storage Service (OSS)	An encrypted and secure cloud storage service that can store, process, and access vast amounts of data from anywhere in the world.
File Store	File Storage NAS	Simple, scalable, on-demand, and reliable network-attached storage (NAS) for use with ECS instances, HPC, and Container Services.
Cloud BigTable	Tablestore	A fully managed Non-Structured Query Language (NoSQL) cloud database service that enables the storage of a massive amount of structured and semi-structured data.

2.3.3 HYBRID CLOUD STORAGE

GCP Product	Alibaba Cloud Product	Description
Storage Transfer Service	Cloud Storage Gateway (CSG)	Cloud Storage Gateway uses OSS for cloud-based storage at the back end and supports standard file and block storage protocols in the industry.





2.3.4 STORAGE DATA SERVICE

GCP Product	Alibaba Cloud Product	Description
Cloud Logging, Pub/Sub, Dataflow	Log Service (SLS)	All-in-one service for log-type data.

2.4 DATABASE

2.4.1 DATA WAREHOUSE

GCP Product	Alibaba Cloud Product	Description
BigQuery	AnalyticDB for MySQL	A real-time data warehousing service that can process PB data with high concurrency and low latency.
	AnalyticDB for PostgreSQL	An online MPP (Massively Parallel Processing) data warehousing service based on the open-source Greenplum Database. It provides an online expansion and performance monitoring service to free your team from complicated MPP cluster O&M.
	Data Lake Analytics (DLA)	An interactive analytics service that utilizes serverless architecture. As a ready-to-use service, DLA does not require any prior setup of infrastructure or upfront management costs. You do not need to maintain instances in DLA, and service is billed based on actual use and needs. DLA uses Structure Query Language (SQL) interfaces to interact with user service clients, which means it complies with standard SQL syntax and provides a variety of similar functions. DLA allows you to retrieve and analyze data from multiple data sources or locations such as OSS and Tablestore for optimal data processing, analytics, and visualization to give better insights and ultimately guide decision-making.



2.4.2 NOSQL DATABASE

GCP Product	Alibaba Cloud Product	Description
Cloud MemoryStore	ApsaraDB for Redis	A fully managed, in-memory database service that is compatible with Redis protocols. It can process data in key-value pairs with low response latency. It is a popular choice for scenarios where high concurrency and low response latency are necessary.
Cloud Firestore/ Firebase	ApsaraDB for MongoDB	A MongoDB-compatible database service that is developed based on the distributed system and high-reliability storage engine of Apsara. It uses a multi-node architecture to ensure high availability and supports elastic scaling, disaster recovery, backup and restoration, and performance optimization.
Cloud Bigtable	ApsaraDB for HBase	A cost-effective cloud-based intelligent NoSQL service that is of high scalability and compatible with the open-source HBase. This service provides core benefits such as low storage costs, high throughput and scalability, and intelligent data processing. It also ensures high availability by deploying data centers across regions.

2.4.3 RELATIONAL DATABASE

GCP Product	Alibaba Cloud Product	Description
Cloud Spanner	PolarDB	A new generation of "Super MySQL, PostgreSQL, or compatible Oracle" supporting up to 100TB storage.
Cloud SQL for MySQL	ApsaraDB RDS for MySQL	Availability and performance enhanced MySQL service supporting versions 5.5, 5.6, 5.7, and 8.0.
Cloud SQL for PostgreSQL	ApsaraDB RDS for PostgreSQL	An on-demand PostgreSQL service with automated monitoring, backup, and disaster recovery capabilities.
Cloud SQL for SQL Server	ApsaraDB RDS for SQL Server	An on-demand database hosting service for SQL Server with automated monitoring, backup, and disaster recovery capabilities.
Oracle on Bare Metal	PolarDB for Oracle	We recommend this migration option with cloud-native features. It has high compatibility with Oracle but with a cost reduction.



2.4.4 UTILITY & TOOLS

GCP Product	Alibaba Cloud Product	Description
Database Migration Service/ Datastream	Data Transmission Service (DTS)	Alibaba Cloud DTS helps you migrate data between data stores, such as relational databases, NoSQL databases, and data warehouses. You can use DTS to migrate your data to Alibaba Cloud or between combinations of cloud and on-premises data systems. DTS supports several data replication modes, including data migration, data integration, data synchronization, and change tracking. You can choose a combination of data replication modes that best suit your use cases.
Cloud Data Fusion/Cloud Composer/ Dataplex/Data Catalog	Data Management Service (DMS)	DMS allows you to manage databases in a centralized manner. In DMS, you can manage relational databases such as MySQL, SQL Server, PostgreSQL, PolarDB for MySQL, PolarDB for PostgreSQL, PolarDB O Edition, PolarDB-X, OceanBase, and Oracle databases. You can also manage online analytical processing (OLAP) databases such as AnalyticDB, DLA, and ClickHouse databases, as well as NoSQL databases such as MongoDB and Redis databases. DMS is a comprehensive data management service that integrates data management, schema management, user authorization, security audit, trend analysis, and data tracking. You can use DMS to manage databases with ease. It enhances data security, improves management efficiency, and maximizes data value.

There are [more database-centric, hands-on cloud solutions for reference](#).



2.5 SECURITY

2.5.1 CLOUD SECURITY

GCP Product	Alibaba Cloud Product	Description
Google Cloud Armor Managed Protection	Anti-DDoS	Comprehensive Distributed Denial of Service (DDoS) protection for enterprises to intelligently defend sophisticated DDoS attacks, reduce business loss risks, and mitigate potential security threats.
Google Cloud Armor Managed Protection	Anti-DDoS Basic	A cloud-based security service that protects your data and application from DDoS attacks.
Google Cloud Armor	Web Application Firewall (WAF)	An easy-to-use service that provides real-time monitoring of servers to ensure high availability.
Certificate Authority Service	Alibaba Cloud SSL Certificates Service	Alibaba Cloud's Secure Sockets Layer (SSL) Certificates Service allows customers to directly apply, purchase, and manage SSL certificates.
Security Command Center	Security Center	A unified security management system that identifies, analyzes, and notifies you of security threats in real-time.
Cloud Asset Inventory	Cloud Config	A configuration audit service that tracks and audits configurations of your Alibaba Cloud resources, helping you to achieve the compliance of resource configurations.
Google Cloud firewalls	Cloud Firewall	Cloud Firewall is one of the first Software as a Service (SaaS) firewalls deployed on the public cloud. It is your first choice for network security that safeguards your businesses in the cloud.
Cloud Audit Logs	ActionTrail	A service that monitors and records the actions of your Alibaba Cloud account, including the access to and use of Alibaba Cloud services using the Alibaba Cloud Management console, calling application programming interfaces (API) operations or Software Development Kits (SDKs).



2.5.2 DATA SECURITY

GCP Product	Alibaba Cloud Product	Description
Cloud Data Loss Prevention	Sensitive Data Discovery and Protection (SDDP)	An all-in-one data security solution that provides various features, such as sensitive data detection, classification, grading, and de-identification, to help you meet compliance requirements specified in General Data Protection Regulation (GDPR) and personal information protection.
Cloud Key Management Service	Key Management Service (KMS)	Easily protect, manage, use, and audit your cryptographic keys on Alibaba Cloud.
Cloud HSM	Data Encryption Service	Industry-standard hardware security modules (HSMs) deployed on Alibaba Cloud.

2.5.3 IDENTITY MANAGEMENT

GCP Product	Alibaba Cloud Product	Description
Identity and Access Management	Resource Access Management (RAM)	Secure your cloud resources with Resource Access Management to define fine-grained access permissions for users and groups.
Cloud Identity	Identify-as-a-Service (IDaaS)	Make identity management a painless experience and eliminate Identity Silos.

2.6 CONTAINER / SERVERLESS / MIDDLEWARE

2.6.1 APPLICATIONS

GCP Product	Alibaba Cloud Product	Description
Cloud Monitoring	Application Real-Time Monitoring Service (ARMS)	Build business monitoring capabilities with a real-time response based on frontend monitoring, application monitoring, and custom business monitoring capabilities.
Cloud Trace	Tracing Analysis	Allows developers to quickly identify root causes and analyze performance bottlenecks for distributed applications.



2.6.2 CONTAINERS

GCP Product	Alibaba Cloud Product	Description
Cloud Run	Elastic Container Instance (ECI), Serverless Container Service	An agile and secure serverless container instance service.
Google Kubernetes Engine (GKE)	Container Service for Kubernetes (ACK)	A Kubernetes-based service that ensures high efficiency for enterprises by running containerized applications on the cloud.
Google Container Registry (GCR)	Container Registry (ACR)	A fully managed cloud-native artifacts management platform. ACR provides end-to-end security, intelligent building, efficient distribution of Docker container images, Helm Charts, and OCI-compatible (Oracle Cloud Infrastructure compatible) artifacts.
Anthos Service Mesh	Alibaba Cloud Service Mesh (ASM)	A fully managed service mesh platform that is compatible with Istio. ASM provides consistent traffic management, secure communication with authentication, and service observability in heterogeneous environments.

2.6.3 MESSAGE QUEUES

GCP Product	Alibaba Cloud Product	Description
Pub/Sub Lite	Message Service	A message queuing and notification service that facilitates the smooth transfer of messages between applications.
Cloud IoT Core	AliwareMQ for IoT	A message service designed for IoT (Internet of Things) and mobile Internet (MI).

2.6.4 MICRO SERVICES

GCP Product	Alibaba Cloud Product	Description
App Engine	Enterprise Distributed Application Service (EDAS)	A Platform as a Service (PaaS) platform for a variety of application deployment options and microservices solutions to help monitor, diagnose, operate, and maintain your applications.

2.6.5 SERVERLESS

GCP Product	Alibaba Cloud Product	Description
Cloud Functions	Function Compute (FC)	A secure and stable, elastically scaled, O&M-free, pay-as-you-go, serverless computing platform.
Workflows	Serverless Workflow	Visualization, O&M-free orchestration, and Coordination of Stateful Application Scenarios.





3. MIGRATION PROCESS

Moving from one cloud provider to another could prove to be a complicated process. Therefore, Alibaba Cloud's methodology has been summarized into five steps based on their experience with thousands of customers, so that you can easily follow the procedure.

1. Assessment

It's advisable to always start with an assessment to understand the system architecture (e.g. server inventory), migration feasibility (e.g. downtime and data requirement), and the potential risk associated with the migration.

2. PoC

This information can then be used for pilot testing, conducting a PoC (Proof of Concept) to ensure that the product functionality can meet the business requirement.

3. Design

Once the PoC has been done, it is time to move to the design stage. After the deployment scheme has been planned, products used in GCP can be mapped to Alibaba Cloud, any potential changes can be made to the application, and plans can be put in place for the required resource size.

4. Implementation

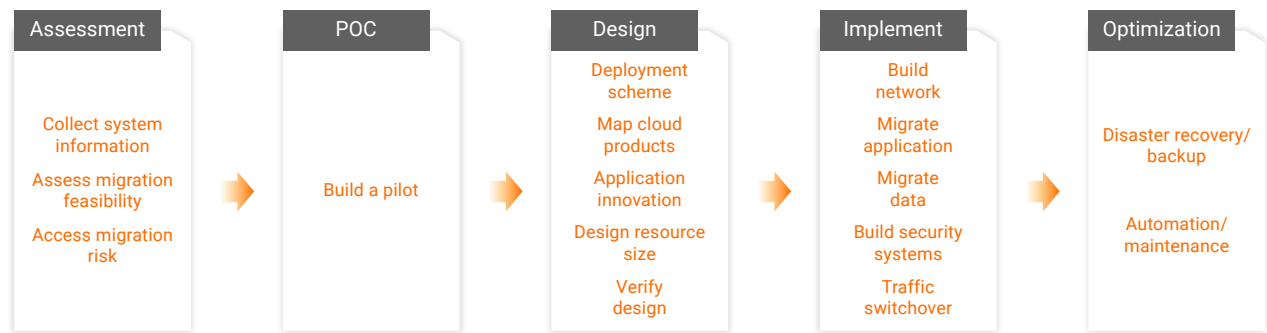
After the solution has been designed, Alibaba Cloud will implement them part by part. First, the network needs to be built, along with the operation environment, and the applications need to be prepared in Alibaba Cloud. Historical data is the first to be moved, whereas transaction data can be ported over during the cutover. More details can be found in the following sessions.

In addition to the implementation plan, a contingency plan should always be prepared in case of rollback.



5. Optimization

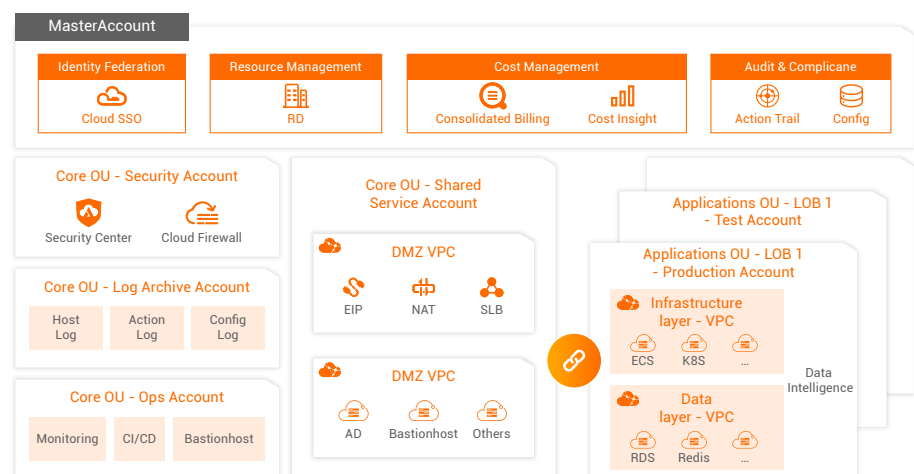
It's always necessary to keep your setup optimized after implementation. Including Disaster Recovery (DR) setup, backup, automation, and maintenance.



4. MIGRATION PRACTICES

4.1 CONSIDER YOUR ACCOUNT STRUCTURE AND IDENTITY AND ACCESS MANAGEMENT (IAM) - LANDING ZONE

In order to ensure the operational safety, compliance, manageability, and scalability of the cloud business, while accelerating the speed of cloud migration, it's recommended that customers design and build an Alibaba Cloud Landing Zone before migrating. When migrating, priority is given to the two neighborhoods of resource planning and identity authority. The typical architecture design is as follows:



4.1.1 MIGRATE MULTI-ACCOUNT STRUCTURE

Alibaba Cloud provides a multi-account structure, which is the basis of LandingZone. The following will explain how you can set up a LandingZone multi-account structure in Alibaba Cloud.

1) DETERMINING THE MASTER ACCOUNT

The first step is to confirm the master account of the enterprise. The enterprise management master account includes the Resource Directory (RD), which is used to manage multiple accounts. You can use this account as a financial custody account and use it to make unified payments and settlements for other accounts, as well as view cloud bills in a unified manner. You can also unify the audit rules configured by this account and send them to each member account.

2) CREATING RESOURCE DIRECTORY

The Alibaba Cloud Resource Directory can provide GCP Manage resources of the same capacity. Component mapping is as follows:

GCP Manage resources	Alibaba Cloud Resource Directory
folder	folder
project	member

You can use the master account to enable the [Resource Directory](#) service and create a resource folder based on the actual service conditions of your enterprise. You can create at least two Resource Directories.

- » **Core:** Centrally manages public service accounts
- » **Custom:** Centrally manages service accounts

3) CREATING A PUBLIC SERVICE ACCOUNT

After the Resource Directory is set up, you need to create three shared service accounts. It should be noted that the functions provided by these public service accounts are realized by different products in Alibaba Cloud:

a) Shared service account:

Corporate shared services will be deployed in this account, such as network configuration, unified access to the public network configuration, etc.

For network-related cloud products, refer to the network section.

b) Log account:

This aggregates the logs of all member accounts, including cloud product operation behavior logs, audit logs, and business logs.

GCP	Alibaba Cloud
Cloud Audit Logs	Action Trail
Cloud Asset Inventory	Cloud Config

c) Security account:

Generally, the security role in the enterprise will use this account to configure related security products. Such as common WAF, high-defense Internet Protocol (IP), etc.

For security-related cloud products, refer to the security section.



4.1.2 ACCESS CONTROL

In Alibaba Cloud, we use products and services such as Resource Access Management (RAM) to achieve identity management and access control capabilities.

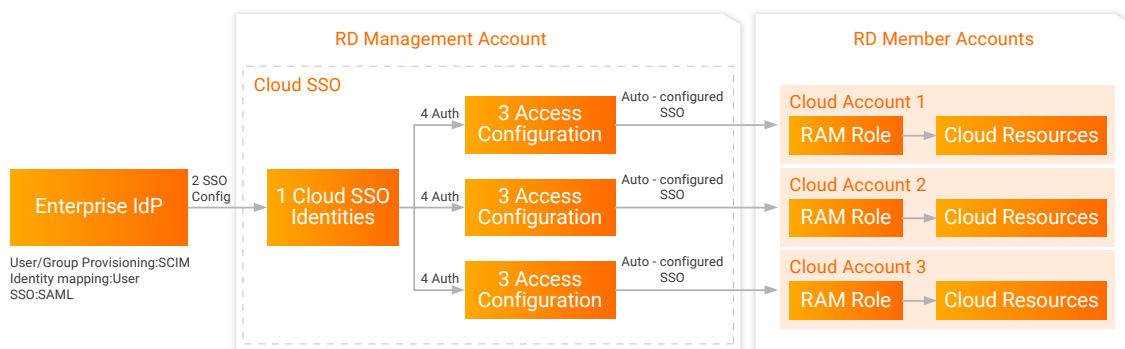
RAM is an advanced access control mechanism based on attribute-based access control (ABAC). When a customer request arrives at Alibaba Cloud, it will evaluate the currently accessed request characteristics, identity characteristics, and resource characteristics, and match them with the configured permissions of the identity to complete the authentication.

The identity and access control granularity supported by each cloud product can be found in cloud services that [support RAM](#) and cloud services that [support Security Token Service \(STS\)](#).

4.1.3 SINGLE SIGN-ON (SSO)

In the multi-account scenario, each account needs to manage its identity permissions and configure the single sign-on between each account and Alibaba Cloud. This management cost is very high.

We recommend using Alibaba Cloud [CloudSSO](#), which can integrate the multi-account unified identity management and access control of the Alibaba Cloud RD.

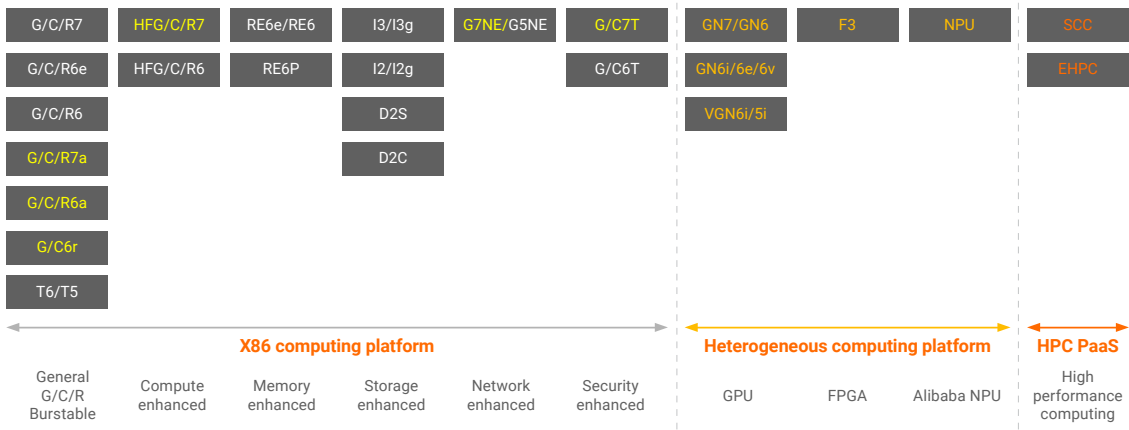


4.2 ELASTIC COMPUTING

4.2.1 CHOOSE ECS INSTANCE TYPE ACCORDING TO GCP VM

Before migrating GCP Virtual Machines (VM) to Alibaba cloud, you have to choose the corresponding instance type carefully. With appropriate instance types, it can improve the application performance while reducing the total cost of computing resources.

Here is the diagram to show the Alibaba Cloud instance families portfolio. You can find the types of workloads or scenarios running on each instance family on the Alibaba Cloud [website](#).



Here are the Alibaba Cloud ECS instance types compared with GCP instance types, which can be used as a brief guideline on how to select instance families.

	Alibaba Cloud	Google
Instance		
Burstable	T6 / T5	e2 shared-core / n1 shared-core (f1/g1)
General Purpose	G6e / G6 / G5	e2-Standard / n2-Standard / n1-standard
Compute Optimized	C6e / C6 / C5	e2-highcpu / n2-highcpu / n1-standard
Memory Optimized	R6e / R6 / R5	e2-highmem / n2-highmem / n1-highmem
Network Optimized	G5ne	N/A
Memory Enhanced	RE6 / RE4	m2-ultramem / m2-megamem / m1-ultramem / m1-megamem
Security Enhanced	C6t / G6t	Confidential VM (AMD CPU only)
High Frequency	HFC7 / HFR7 / HFG7 / HFC6 / HFR6 / HFG6	c2-standard
Instance with Local SSD	I3 / I3g / I2ne / I2gne	N1 / N2 / N2D / C2
Instance with Local HDD	D2S / D2C	N/A
GPU Instances	GPU / FPGA	a2-highgpu / a2-megagpu / n1 (able to add vGPU)
AMD Instances	G6a / C6a / R6a	N2D / E2
ARM Instances	G6r	N/A
Instance Forms		
Virtual Machine	VM	VM
Dedicated Hosts	Dedicated Hosts	Sole-tenant Nodes
Bare Metal	Bare Metal	Bare Metal
Serverless Container	Elastic Container Instances	Cloud Run

	Alibaba Cloud	Google
Supporting Products / Features		
Deployment Set	Deployment Set	Instance Groups
Auto Scaling	Autoscaling (ESS)	Compute Engine Autoscaler
Automated Provision	Resource Orchestration Service (ROS)	Cloud Deployment Manager
Automated Ops & Maintenance	Ops Orchestration Service (OOS)	Cloud Operation
Migration Tool	Server Migration Center	Migrate for Compute Engine
Purchasing Model		
Subscription Model	Month / Year (not recommended moving forward)	**Committed use discounts
On Demand	Pay as you go	Pay as you go
Reserved Instances	Reserved Instances	N/A
Saving Plan	Saving Plan	N/A
Preemptible Instances	Preemptible	Preemptible Machines
SAAS Products		
Cloud Desktops	Elastic Desktop Service	Virtual Desktops (3 rd Party Solution)
Bundled Products		
Simple Application Server	Simple Application Server	N/A

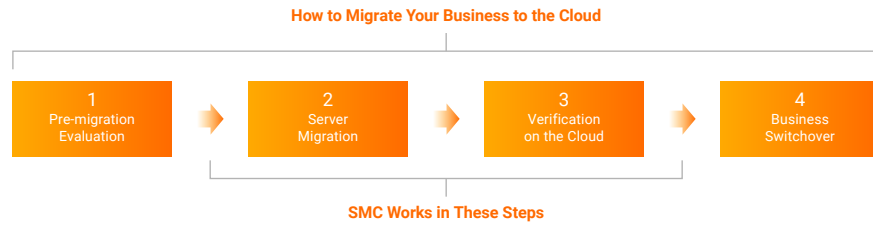
4.2.2 USING SMC TO MIGRATE VM FROM GCP TO ALIBABA CLOUD

WHAT IS SMC?

Alibaba Cloud Server Migration Center (SMC) is a free tool for Cloud Migration, which was launched in November 2017. SMC allows you to migrate one or more source servers to Alibaba Cloud. Source servers can be servers in data centers, VMs, servers on other cloud platforms, and other types of servers. SMC helps you automate the migration of server application environments and simplifies the migration of system images to ECS instances. To facilitate the management of migration tasks, SMC provides service-oriented API operations that allow you to create and execute migration scripts and query the migration progress.

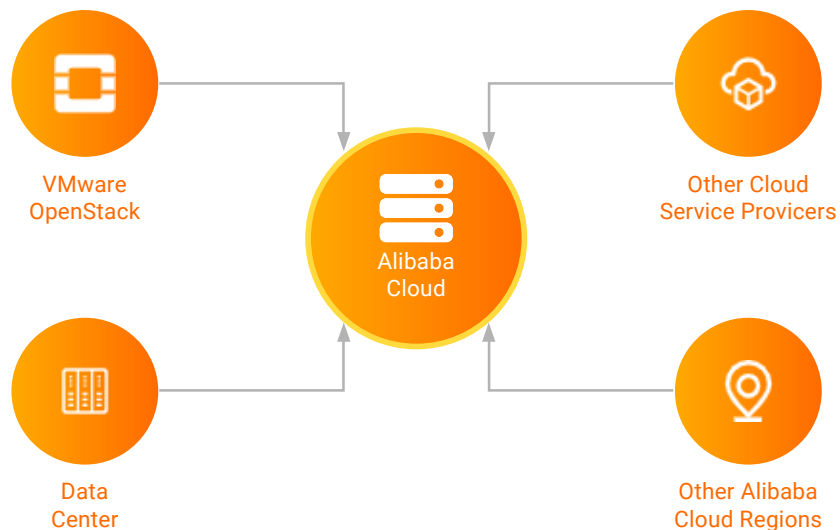


HOW SMC CAN HELP?



SMC works in server migration and can help to verify VM after migration. Before SMC migrates VM, application pre-migration evaluation should be completed. Migrating VM using the pay-as-you-go (PAYG) purchasing model before the business shift to Alibaba Cloud is the suggested route and, after that, you can change PAYG to other purchasing models, such as monthly/yearly subscriptions for those all-time running workloads.

Notes: for database migration, DTS is recommended



SMC allows you to migrate servers from your own data centers, your local VMs (VMware, Virtual Box, Xen, and KVM), and other cloud platforms, such as Amazon Web Services (AWS), Azure, Google Cloud, Huawei Cloud, Tencent Cloud, UCloud, Telecom Cloud, and QingCloud.

KEY BENEFITS OF SMC?



Zero Dependency on Underlying Environment

SMC supports both P2V and V2V, multiple file system formats, and multiple disk types.



Zero Local Storage

Data is directly migrated to the cloud and does not consume local storage.



Zero Downtime

The migration does not affect the services of the source system and does not cause downtime.



Lightweight and Portable

The SMC client is small and does not require installation. It involves simple deployment.



Flexible Configuration

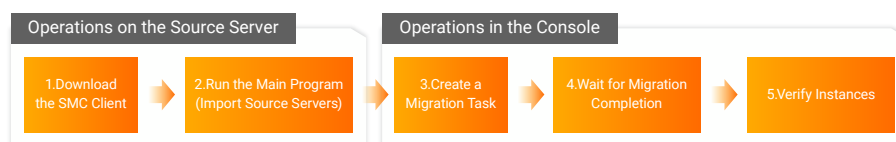
SMC provides a variety of migration and transmission solutions that you can choose from.



One-click Migration

The migration is managed. You can start migration with a few clicks.

SMC MIGRATION STEPS



Before you begin SMC migration, the following things need to be well-prepared:

- » [RAM and AccessKey are prepared](#)
- » Network connection (either public or private network) between source VM and destination VM.
- » Destination network environments created including VPC, vSwitch, and security group.

There are five steps to migrate a VM to Alibaba Cloud:

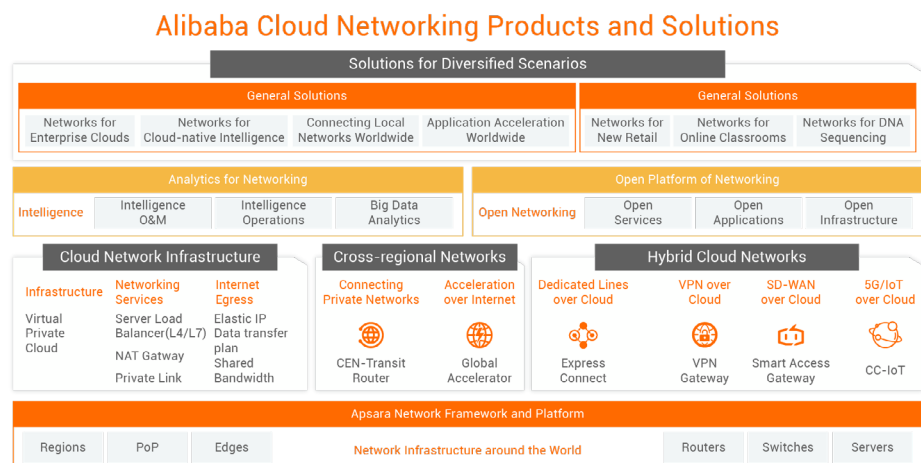
1. Download and install the SMC client on your source server.
 - » The client has 32-bit and 64-bit versions for Windows and Linux, please select the right one.
 - » `"go2aliyun_client"` is the main program.
 - » `"user_config.json"` is the user configuration file.
2. Run the main program in the source server to import the source server information to Alibaba Cloud SMC service.
3. Create a migration task in the Alibaba Cloud SMC console and begin the job.
4. Migration will run automatically until completed; the time spent depends on the source server data capacity and outbound bandwidth.
5. Verify the migration result.

Here are some useful links for SMC tools. If you have any trouble using this tool, you can contact the Alibaba Cloud Global Technology Service (GTS) team for support:

- » [SMC Best Practice](#)
- » [SMC Console](#)
- » [SMC Help](#)

4.3 NETWORKING

Alibaba Cloud's networking services provides you with a highly stable, low latency, and high-speed network with flexible hybrid cloud connections. The networking services deliver secure and reliable communication to and from International Data Corporation's (IDC's) in 25 regions globally. With more than 80 areas deployed globally, Alibaba Cloud's large node network gives you access to close-by nodes, shortening geographic distances.



CROSS-REGION GLOBAL NETWORK CONNECTION

Alibaba Cloud provides fast and cost-effective networking services for multinational enterprises to build cross-region network connections. It ensures that your cross-region interconnections are legal and compliant.

COMPREHENSIVE HYBRID NETWORK SOLUTION

For any industry, Alibaba Cloud's hybrid cloud network solution enables you to connect your on-premise data from IDC, headquarters, branches, or offline stores to Alibaba Cloud. The hybrid network connections ensure that your cross-region interconnections are secure, reliable, and flexible.

QUICKLY AND EASILY SET-UP AND MANAGE YOUR NETWORKS

Alibaba Cloud provides you with an easy-to-use interface to manage your networking services, such as CEN. You can set up a multi-region networking environment in four steps within five minutes.

4.3.1 SLB

CLB deployment

ALB deployment

Differences between ALB and CLB

Type	ALB	CLB
Service Scope	» Robust processing capabilities at Layer 7 and advanced routing features » Supports protocols such as HTTP, HTTPS, and QUIC at the application layer » Application-oriented	» Robust processing capabilities at Layer 4 » Supports protocols such as TCP, UDP, HTTP, and HTTPS » Network-orientated
Architecture and performance	» Based on the network function virtualization (NFV) platform and supports auto scaling » One million QPS per instance	» Developed on top of a physical machine architecture » Five million QPS per instance
Forwarding capabilities	» Supports content-based routing » Supports HTTP rewrites, redirects, overwrites, and throttling	» Basic features at Layer 7 » Supports only domain name-based and URL -based forwarding
Operations and maintenance	» Supports auto scaling » Supports auto scaling to withstand traffic spikes	» Provides different specifications » Requires an estimate of traffic spikes » Requires manual upgrades of specifications
Cloud-native support	» Serves as a cloud-native gateway » Supports traffic splitting, mirroring, canary releases, and blue-green deployments	» Must be used in combination with container services such as Container Service for Kubernetes (ACK) and Serverless Kubernetes (ASK)
Scenarios	» Supports high-performance auto scaling for Internet application at Layer 7 » Maintains low network latency for audio and video streaming applications in high traffic scenarios » Supports canary releases and blue-green deployments for cloud-native applications	» Ensures high reliability when CLB distributes requests to websites and systems at Layer 4 » Supports high concurrency and ensures high performance in traffic distribution » Supports zone-disaster recover and cross-region disaster recovery

Differences with GCP CLB

- » GCP CLB does not support configuring Transmission Control Protocol (TCP)/User Datagram Protocol (UDP) listening, whereas Alibaba Cloud can support the creation of TCP/UDP listening. TCP allows you to transmit data in a reliable and accurate manner but at relatively low speeds. This means that you can use TCP to transfer files, send or receive emails, and perform remote logins. You can also add a TCP listener to forward TCP requests. UDP applies to services that prioritize real-time content delivery over reliability, such as video conferencing and real-time

quote services.

- » GCP CLB does not support client certificates. The client certificate is used to authenticate the identity of a client. A server authenticates the identity of a client by verifying the certificate sent by the client. You can sign a client certificate with a self-signed CA (Certificate Authority) on Alibaba Cloud ALB.

4.3.2 VPC

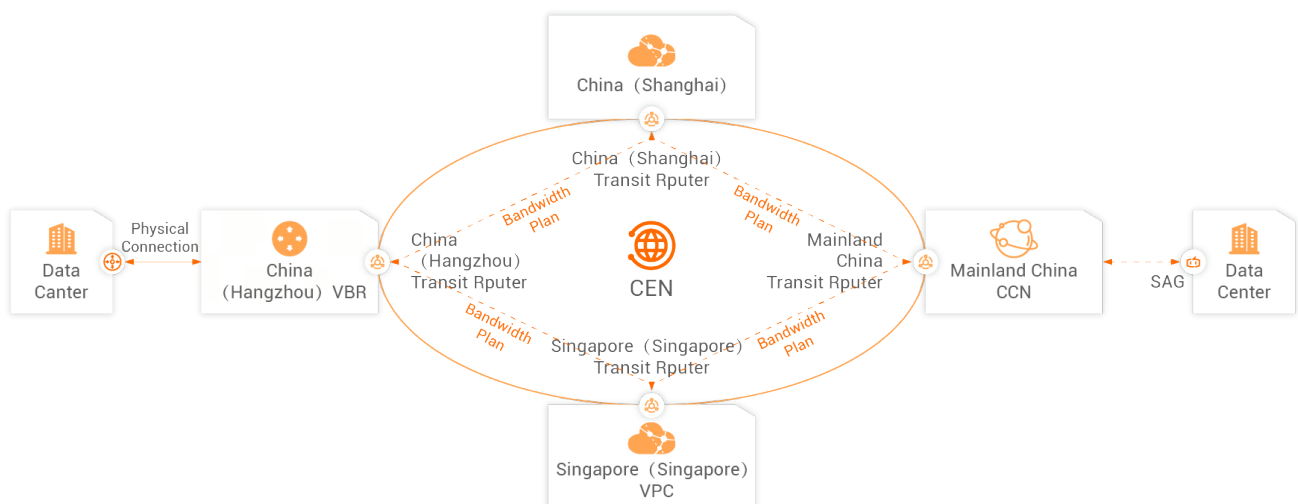
VPC deployment

Differences with GCP VPC

- » On GCP, you can use a single VPC to span multiple regions without communicating across the public internet. On Alibaba Cloud, you need to deploy CEN to span multiple region VPCs.
- » GCP VPC does not support HAVIP (High Availability Virtual IP). You can use HAVIPs with high-availability (HA) software, such as Keepalived, to deploy services in active/standby mode on Alibaba Cloud. This improves the availability of your services.

Cloud Enterprise Network (CEN)

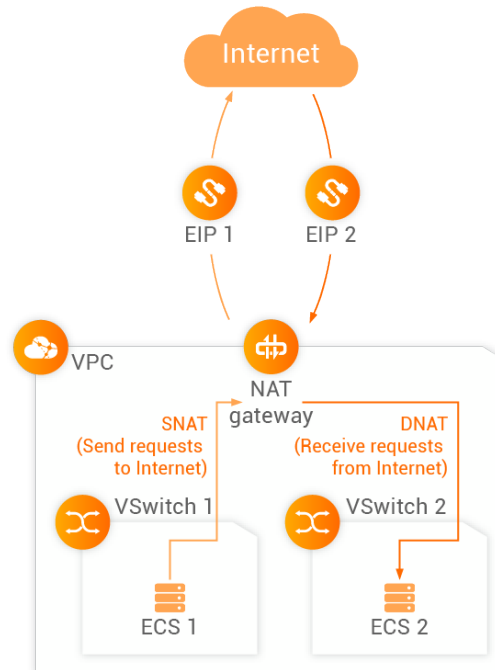
- » CEN can implement network communication in the same region or across regions by establishing private connections between VPCs or between VPCs and data centers. In addition, CEN supports custom communication, isolation, and redirect policies within a region. This allows you to establish a global enterprise-class private network that is flexible, reliable, and large-scale.



4.3.3 NAT GATEWAY

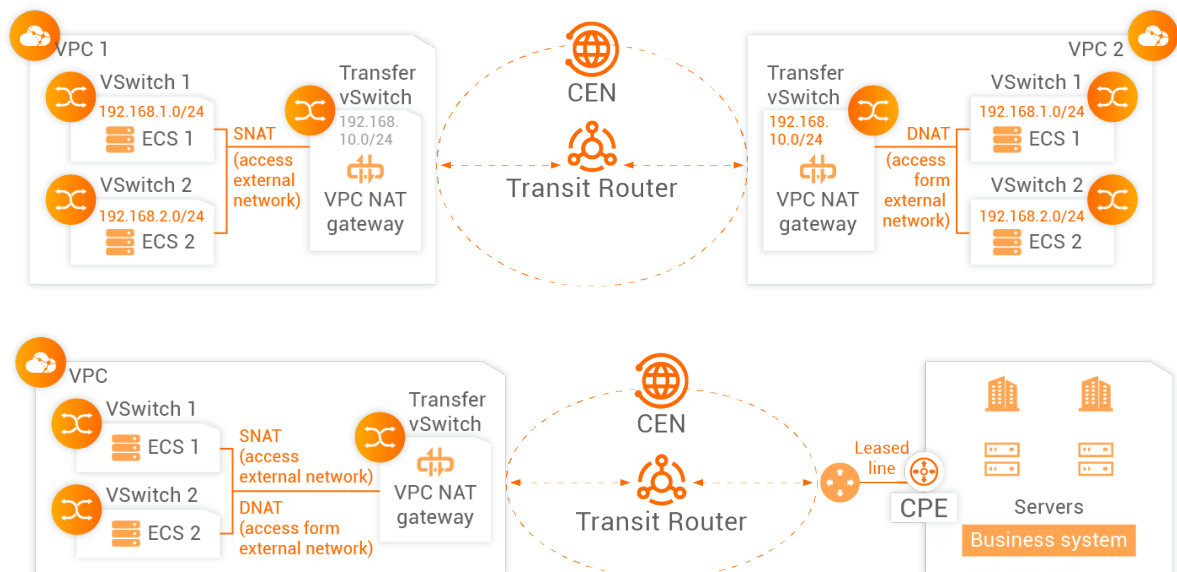
Internet NAT gateway deployment

- » Internet NAT gateways are enterprise-class gateways that provide the SNAT and DNAT features. A NAT gateway can provide a throughput capacity of up to 100 Gb/s. NAT gateways also support cross-zone DR.



VPC NAT gateway deployment

- » VPC NAT gateways provide NAT services to ECS instances in a VPC. The ECS instances can use the NAT IP addresses to access your data center or other VPCs, as well as provide services to external networks.



Differences with GCP Cloud NAT

- » GCP Cloud NAT cannot support SNAT and DNAT from other VPCs while Alibaba Cloud NATGW (NAT gateway) does support this scenario. This allows Alibaba Cloud NATGW to accept VPC ingress/egress traffic from other VPCs.
- » GCP Cloud NAT can only bind one Elastic IP address (EIP), while Alibaba Cloud NATGW can bind multiple Elastic IP addresses. This allows you to create different SNAT rules that map different egress traffic by sourcing Classless Inter-Domain Routing (CIDR) blocks to different EIP.
- » You can build VPC NAT on Alibaba Cloud. VPC NAT gateway allows you to create custom SNAT and DNAT entries to translate private IP addresses, while GCP Cloud NAT doesn't have such a configuration.

4.3.4 BUILD A PRIVATE CONNECTION BETWEEN ALIBABA CLOUD AND GCP

VPN Gateway

VPN Gateway is an Internet-based service that securely and reliably connects enterprise data centers, office networks, and third-party public cloud vendors like AWS, Azure, or GCP to VPCs of Alibaba Cloud through encrypted channels.

Configurations:

- » [Alibaba Cloud side](#)
- » [GCP side](#)

Express Connect

Express Connect can work in an Active/Active or Active/Standby manner. Both Alibaba Cloud and GCP support the feature.

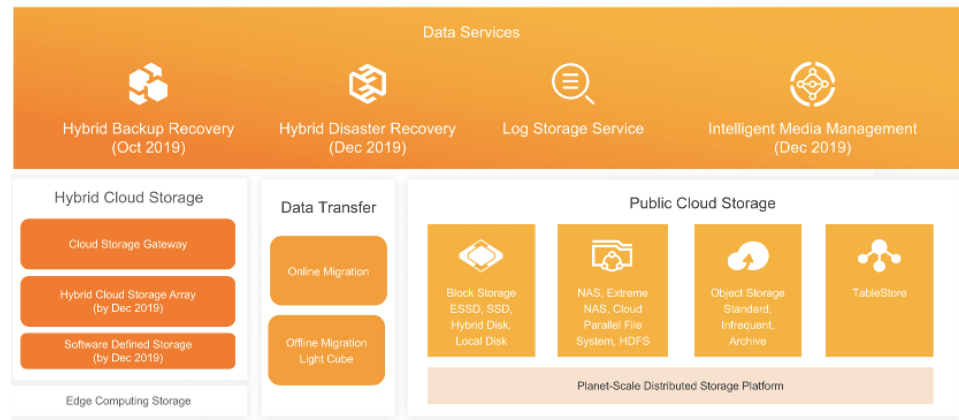
Configurations:

- » Alibaba Cloud: [Active/Active](#) and [Active/Standby](#)
- » [GCP](#)



4.4 STORAGE

For storage, Alibaba Cloud provides many products and services, as can be seen below. The first step to consider is data migration for the foundational storage products.



4.4.1 DATA MIGRATION

When migrating from GCP, your data might come from the following sources:

Source	Destination
Google Cloud Storage	OSS
Files in File Store	File Server NAS

When you migrate your data from GCP to Alibaba Cloud or perform any cross-cloud data migration in general, you need to consider the following things:

DATA SIZE AND TIMELINE

If the total data size is within a few TBs and the migration task is not time-sensitive, consider using online migration through the Internet. If the total data size is 100TB or above, consider using online migration through the express route. For example, the table below shows sample figures to see how long a migration job can take for reference. In real scenarios, the time to migrate also depends on the actual bandwidth and number of items to be migrated.

Data to migrate	Bandwidth	Time to migrate
1 TB	1Gb/s (Internet)	230 minutes
1 TB	10Gb/s (Internet)	23 minutes
1 TB	100Gb/s (express route)	2.3 minutes
1 PB	1Gb/s (Internet)	100 days
1 PB	10Gb/s (Internet)	10 days
1 PB	100Gb/s (express route)	1 day



COST

The cost for data migration includes the following:

Fees	Description
Internet Egress fee from GCP	If you choose approach 1, GCP charges the Internet egress fee.
Dedicated Interconnect Egress from GCP	If you choose approach 2, GCP charges the egress fee for direct connect.
API request fee from GCP	AWS charges API requests that will be used to retrieval the data from their S3 (cloud object storage).

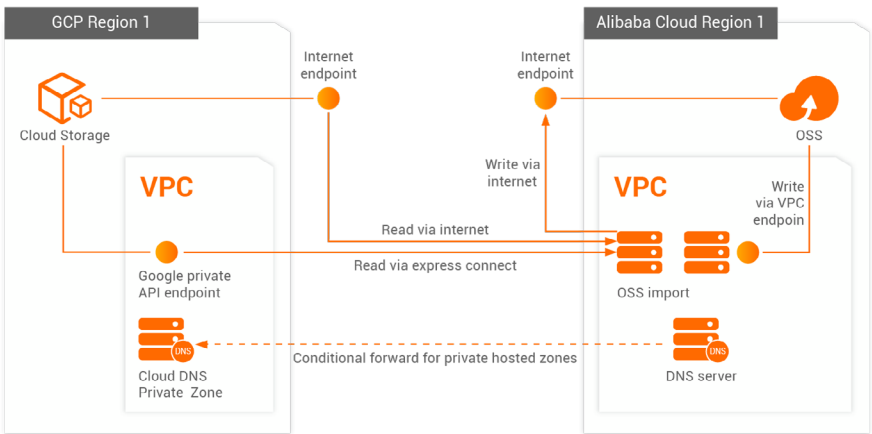
For example, the egress fee to the Internet is as below.

Monthly Usage	Egress to Worldwide Destinations (excluding Asia & Australia) (per GB)	Egress to Asia Destinations (excluding China, but including Hong Kong) (per GB)	Egress to China Destinations (excluding Hong Kong) (per GB)	Egress to Australia Destinations and Egress from Cloud Storage regions located in Australia (per GB)	Ingress
0–1 TB	\$0.12	\$0.12	\$0.23	\$0.19	Free
1–10 TB	\$0.11	\$0.11	\$0.22	\$0.18	Free
10+ TB	\$0.08	\$0.08	\$0.20	\$0.15	Free

CROSS-REGION MIGRATION

If you plan to migrate to another region, it is recommended to use a close-by region as an intermediate region and then perform cross-region replication for the initial migration.

The high-level migration approach is illustrated below:





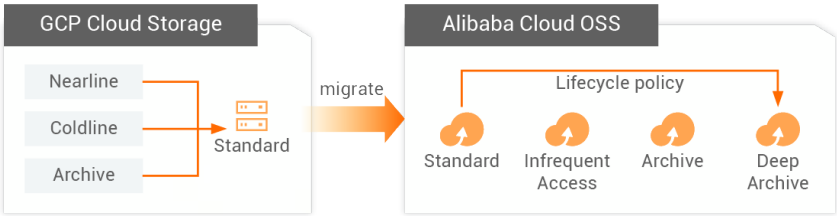
1) Migration Data from GCP Cloud Storage to OSS

Choose the right storage class

Below is a mapping between different storage classes of S3 and OSS. Choose the right storage class:

S3 Storage Class	OSS Cloud Storage Class
Standard Storage	Standard
Nearline Storage	Infrequent Access
Coldline Storage	Archive
Archive Storage	Cold Archive

Since the Standard tier is optimized for frequent read and write, data migration should be completed using the Standard tier to optimize for the migration cost.



Migration from GCP Cloud Storage Standard to OSS via the Internet

The online migration feature in [Data Transport](#) is a serverless migration engine that natively allows you to migrate your GCP Cloud Storage to OSS. Use this tool only when you do online migration via the Internet. The tool allows you to create GCP buckets and OSS as target addresses and then create migration jobs between the two targets.

Migration from GCP Cloud Storage Standard to OSS via express connect

[ossimport](#) is a tool used to migrate data to OSS. You can deploy ossimport on local servers or ECS instances in the cloud to migrate data stored locally or in other cloud storage systems to OSS. It supports standalone mode (for <30TB migration) and distributed mode for (>30TB migration). To be noted, when ossimport read data from Google Cloud Storage, you need to configure conditional Domain Name System (DNS) forwarding for private hosted zones. Refer to the "on-premises" section in [this document](#) for more details.



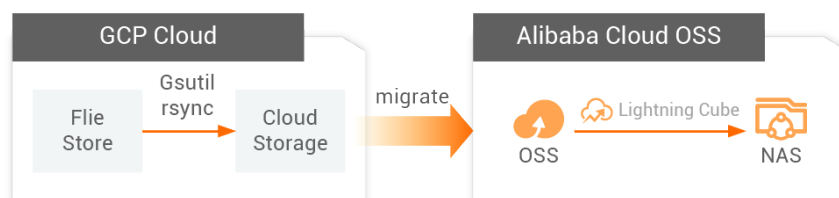
Migration from Neoline, Cloudline, and Archive Storage to OSS

This is important if you plan to migrate from Glacier Archive Tier or Glacier Deep Archive Tier to OSS Archive or OSS Cold Archive. Since data retrieval from the archive tier incurs significant costs, it is recommended to convert to the Standard tier and then follow the Standard to OSS approach for data migration.

2) Migration from File Store to NAS

Migration from File Store to NAS leverages Cloud Storage to OSS migration:

- » [Use gsutil rsync to transfer files from File Store to Cloud Storage.](#)
- » Migrate Google Cloud Storage to OSS as described in the previous section.
- » Use online migration to [transfer OSS to NAS](#).



4.4.2 APPLICATION MIGRATION

For object storage, you need to switch to using OSS SDK. Though both Google Cloud Storage API and OSS API are S3 compatible, they do not interop with each other.

During the transition, your previous application might still be generating data and writing to Cloud Storage. Your new application based on OSS needs to be able to read data that is newly generated. The back-to-origin setting allows you to retrieve the data from Cloud Storage when the data is not available in OSS. Follow [this document](#) for more details.

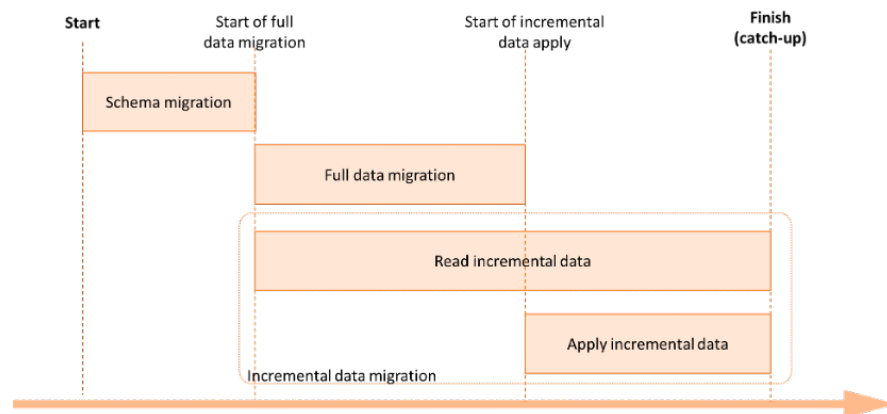
4.5 DATABASE

Database migration is always a cornerstone for cloud migration, either from on-premise to public cloud or from one cloud platform to another cloud platform. The database scheme is clear and features sophisticated migration techniques due to data reliability, high availability, application compatibility, and business continuity. For the overview of the database migration to Alibaba Cloud, please refer to [this webpage](#).



Specifically, most of the cases are covered when migrating databases from Google Cloud to Alibaba Cloud. Alibaba Cloud lists the migration guide according to the types of databases below. Even for the same type of database, there may be more than one kind of alternative database cloud service as the target database on Alibaba Cloud.

Before heading into the database migration guide, let's talk more about [DTS](#), which is the key cloud service working as the data pipeline between the source and target databases, which also works in [major migration scenarios](#). There are generally three task types in DTS for migration:



Migration type	Description
Schema migration	DTS migrates the schemas of the required objects from the source database to the destination database. Tables, views, triggers, and stored procedures can be migrated. For schema migration between heterogeneous databases, DTS converts the schema syntax based on the syntax of the source and destination databases. For example, it converts the NUMBER data type in Oracle databases into the DECIMAL data type in MySQL databases.
Full data migration	DTS migrates historical data of the required objects from the source database to the destination database. If you select only the Schema Migration and Full Data Migration as the migration types, DTS does not migrate incremental data generated in the source database. <i>Note: To ensure data consistency, Alibaba Cloud recommends that you do not write data to the source database during Full Data Migration. To ensure service continuity, you must select Schema Migration, Full Data Migration, and Incremental Data Migration when you configure a data migration task.</i>
Incremental data migration	DTS retrieves static snapshots that are used for Full Data Migration from the source database and migrates the snapshot data to the destination database. Then, DTS synchronizes incremental data generated in the source database to the destination database. <i>Note During Incremental Data Migration, data is synchronized between the source and destination databases in real-time. The migration task does not automatically stop. You must manually stop the migration task.</i>

4.5.1 GENERAL STEPS

Usually, there are applications running on top of the target database system on Google Cloud. Before we take the actual migration actions, here are the general steps for most cases.

1. Target database planning and selection

Read through the following questions to determine the target database on Alibaba Cloud:

Which database to use on Alibaba Cloud?

- » Cloud-hosted Alibaba Cloud Database vs Self-built database on ECS. Please refer to [ApsaraDB for RDS vs. Self-built Databases](#) for a brief comparison. Alibaba Cloud recommends the [cloud-hosted Alibaba Cloud Database](#) for better reliability, serviceability, cost-efficiency, upstream and downstream cloud service integration, etc.
- » Alibaba Cloud Database Options. In case there are multiple options for a single type of database migration (e.g., RDS MySQL or PolarDB for MySQL in section 2.4.1 MySQL migration, PolarDB for Oracle, RDS MySQL, PolarDB for MySQL, or AnalyticDB for PostgreSQL in section 2.4.4 Oracle migration), the key differences will be elaborated in each section.

Which region, edition, and specification to use?

- » Please refer to the product purchase pages for database product availability in global regions.

Database Service and relevant Global Region and Purchase Page

ApsaraDB RDS	TSDB (Time Series Database)
PolarDB	InfluxDB
Redis	AnalyticDB MySQL
MongoDB	AnalyticDB PostgreSQL
HBase	ClickHouse

2. Network connectivity setup

DTS is the key service for the actual migration procedure. So the network connectivity is to make sure the DTS could connect to the target database on Google Cloud. DTS itself is a cloud service on Alibaba Cloud. After the target database has been determined and planned, it is recommended to create a DTS migration task at the same region with the target database on Alibaba Cloud (please refer to [this page](#) for creating DTS migration tasks in global regions).



Make sure the network connectivity from Alibaba Cloud to the source Google Cloud is working either through the public IP: Port for database connection from DTS to the source database, or the network is connected via CEN, Express Connect, VPN Gateway, or Smart Access Gateway.

Refer to [this document](#) to add the CIDR blocks of DTS servers at a specific region (where the DTS migration task is located) to the security settings of the source Google Cloud database.

3. Database migration

When the prerequisites are ready, please refer to the following sections for the actual migration procedure. It's important to take note of the connection status and data transmission performance during the DTS Full Data Migration and Incremental Data Migration. Please refer to the following documents:

- » [The connection status and performance of Full Data Migration](#)
- » [The connection status and performance of Incremental Data Migration](#)

4. Application switch over

When the target database is running, the database connection Uniform Resource Identifier (URI) needs to be updated in the application to switch over to the new database on Alibaba Cloud.

4.5.2 MYSQL

Source database on Google Cloud	Target database on Alibaba Cloud	Migration guide
Self-built MySQL on Compute Engine	ApsaraDB RDS for MySQL This is the most common migration scenario with 100% MySQL compatibility.	» Migration guide
	PolarDB for MySQL Migrate to cloud-native MySQL with computing and storage decoupled architecture to gain fast elasticity and up to 100TB data storage.	» Migration guide
Cloud SQL for MySQL	ApsaraDB RDS for MySQL This is the most common migration scenario.	» Migration guide
	PolarDB for MySQL Migrate to cloud-native MySQL with computing and storage decoupled architecture to gain fast elasticity and up to 100TB data storage.	<i>Before migration procedure, make sure the connectivity to Cloud SQL for MySQL works well.</i> » Migration guide

4.5.3 POSTGRESQL

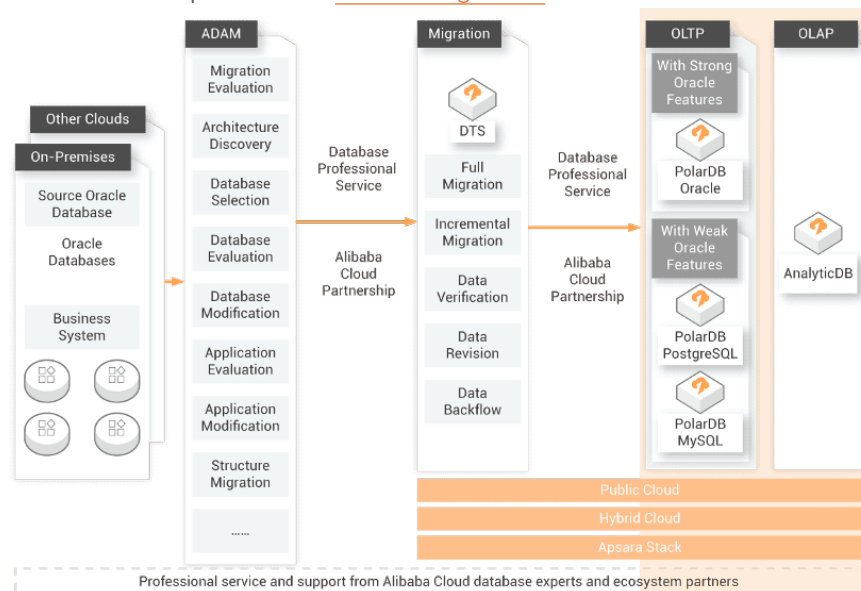
Source database on Google Cloud	Target database on Alibaba Cloud	Migration guide
Self-built PostgreSQL on Compute Engine	ApsaraDB RDS for PostgreSQL	» Full Data Migration » Incremental Data Migration for PostgreSQL versions 9.4.8 to 10.0 » Incremental Data Migration for PostgreSQL versions 10.1 to 13
Cloud SQL for PostgreSQL	ApsaraDB RDS for PostgreSQL	Before migration procedure, make sure the connectivity to Cloud SQL for MySQL works well. » Full Data Migration » Incremental Data Migration for PostgreSQL versions 9.4.8 to 10.0 » Incremental Data Migration for PostgreSQL versions 10.1 to 13

4.5.4 SQL SERVER

Source database on Amazon	Target database on Alibaba Cloud	Migration guide
Self-built SQL Server on Compute Engine	Cloud SQL for SQL Server	» Full Data Migration » Incremental Data Migration
Cloud SQL for SQL Server	ApsaraDB RDS for PostgreSQL	Before migration procedure, make sure the connectivity to Cloud SQL for SQL Server works well. » Full Data Migration » Incremental Data Migration

4.5.5 ORACLE

It would take more time for migrating Oracle from Google Cloud to cloud-native databases on Alibaba Cloud. That's why Alibaba Cloud provides a comprehensive [Oracle Migration](#) on Alibaba Cloud.



Source database on Google Cloud	Target database on Alibaba Cloud	Migration guide
Self-built Oracle on Compute Engine	PolarDB for Oracle	» Migration guide
	Alibaba Cloud recommends this migration option with cloud-native features, high compatibility with Oracle but with cost reduction.	
	ApsaraDB RDS for MySQL	» Migration guide
	If you decide to come to the MySQL ecosystem with strategic consideration in the long term and to avoid vendor lock-in.	
	PolarDB for MySQL	» Migration guide
	If you decide to come to MySQL ecosystem (with cloud-native MySQL on Alibaba Cloud) with strategic consideration in the long term and to avoid vendor lock-in.	
	AnalyticDB for PostgreSQL	» Data migration
	Migrate from Oracle Real Application Clusters (RAC) or Oracle data warehouse in most cases.	» Application migration
	Message Queue Instance: Alibaba Cloud Kafka	» Migration guide
	Connect to Oracle on Google Cloud and get full data and incremental data to the message queue. Then the downstream application or data system can consume message data from Kafka.	

4.5.6 IBM DB2

Source database on Google Cloud	Target database on Alibaba Cloud	Migration guide
Self-built IBM Db2 on Compute Engine	ApsaraDB RDS for MySQL	» Migration guide
	If you decide to come to the MySQL ecosystem with strategic consideration in the long term and to avoid vendor lock-in.	



4.5.7 MONGODB ATLAS

Source database on Google Cloud	Target database on Alibaba Cloud	Migration guide
MongoDB Atlas on Google Cloud	ApsaraDB for MongoDB	» Migration using DTS » Migrate data from MongoDB Atlas to ApsaraDB for MongoDB by using mongodump and mongorestore

4.5.8 MEMORYSTORE FOR REDIS/REDIS

Source database on Google Cloud	Target database on Alibaba Cloud	Migration guide
Memorystore for Redis	ApsaraDB for Redis	» Migration guide
Self-built Redis on Compute Engine	ApsaraDB for Redis	» Migration using DTS » Migration using Redis-Shake (within the guide, follow the similar operation on Google Cloud Compute Engine just like on Alibaba Cloud ECS)

4.5.9 BIGQUERY

Source database on Google Cloud	Target database on Alibaba Cloud	Migration guide
BigQuery	MaxCompute	» Migration guide
BigQuery	AnalyticDB for MySQL	» Step 1: migrate from BigQuery to MaxCompute » Step 2: migrate from MaxCompute project to RDS for MySQL instance » Step 3: synchronize data from a RDS for MySQL instance to an AnalyticDB for MySQL cluster

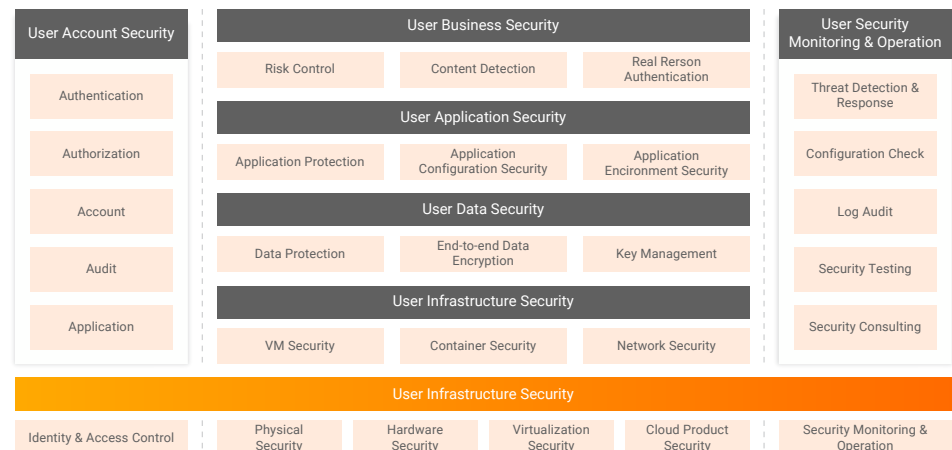
4.5.10 GREENPLUM

Source database on Google Cloud	Target database on Alibaba Cloud	Migration guide
Self-built Greenplum on Compute Engine	AnalyticDB PostgreSQL	» Migration guide



4.6 SECURITY

4.6.1 ALIBABA CLOUD SECURITY ARCHITECTURE

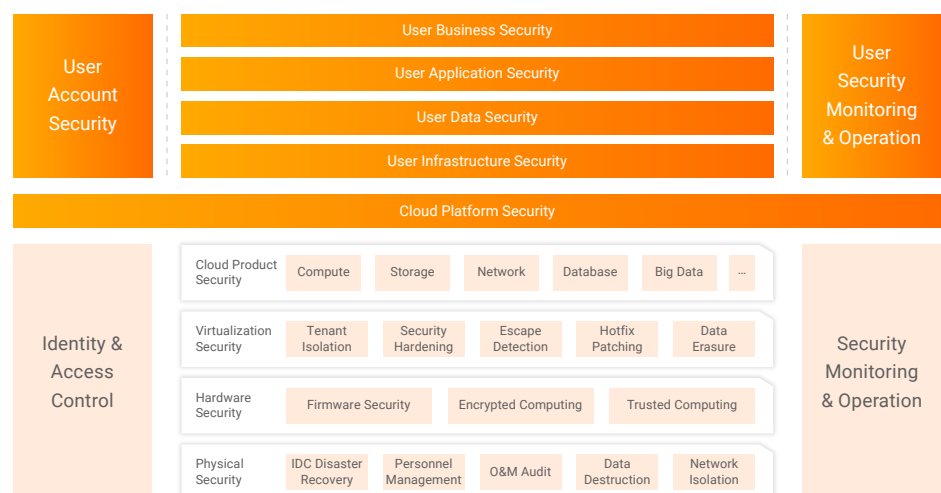


Alibaba Cloud provides a security architecture that offers protection in five horizontal dimensions and two vertical dimensions. The two vertical dimensions include account security (identity and access control) and security monitoring and operations. Note that these vertical dimensions include different implementations for both user security and cloud platform security.

The five horizontal dimensions include cloud platform security at the bottom layer, as well as infrastructure security, data security, application security, and business security for cloud users.

This chapter introduces the overall security architecture and describes the key features of each layer that covers various Alibaba Cloud products and security services.

CLOUD PLATFORM SECURITY

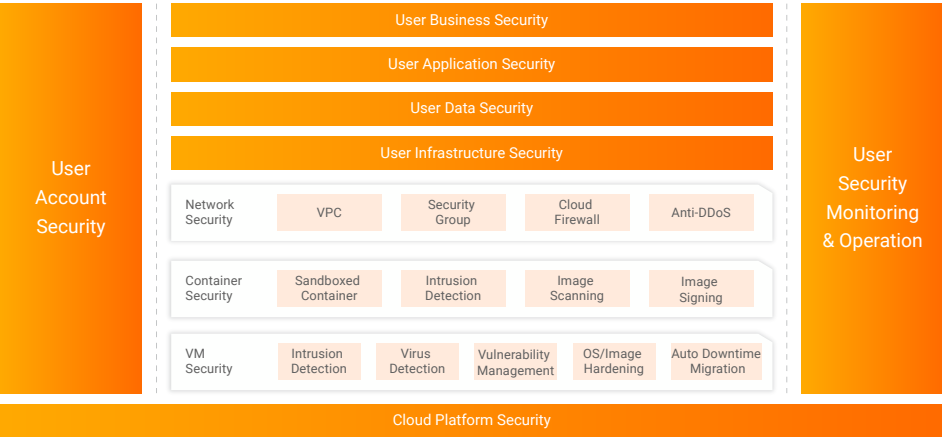


The cloud platform security architecture includes the basic security capabilities provided by the Alibaba Cloud platform. In the two vertical dimensions, identity and access control and security monitoring and operations are invisible to customers.

Similarly, customers can enjoy high-level protection capabilities of Alibaba Cloud without performing configurations at the physical security, hardware security, and virtualization security layers.

The cloud product security layer includes the security capabilities provided by the cloud platform for customers. Some capabilities such as tenant isolation are enabled by default. Some capabilities such as data encryption must be enabled and properly configured by customers to function properly.

USER INFRASTRUCTURE SECURITY

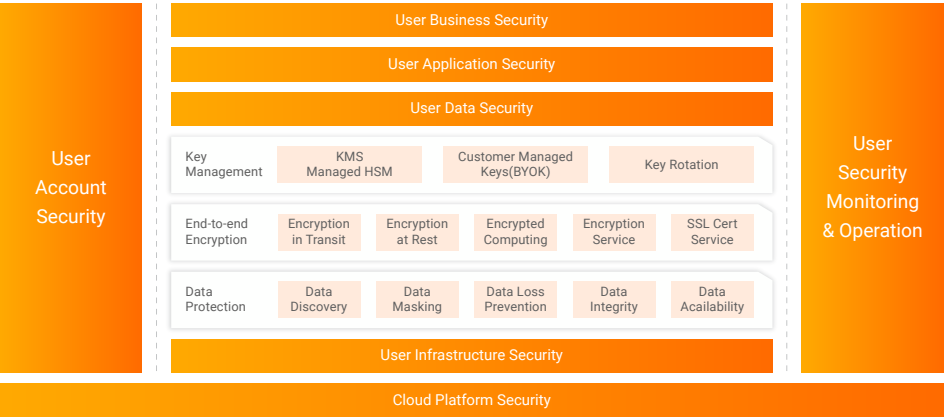


The user infrastructure security capabilities and requirements of cloud users focuses on three aspects: host security, container security, and network security. These aspects involve the use of networks to isolate and protect the most important computing resources. Cloud applications and services of users are directly or indirectly built (through upper-layer cloud services) and run on the basic computing and network service modules.

In addition to computing and network resources, storage resources are also crucial to users, especially in the protection of user data. The user-side security architecture also includes two vertical dimensions: user account security and user security monitoring and operations. These vertical dimensions are involved in each horizontal dimension, such as configuration security checks for VM hosts during security monitoring and control over permissions of various cloud resources for account security.

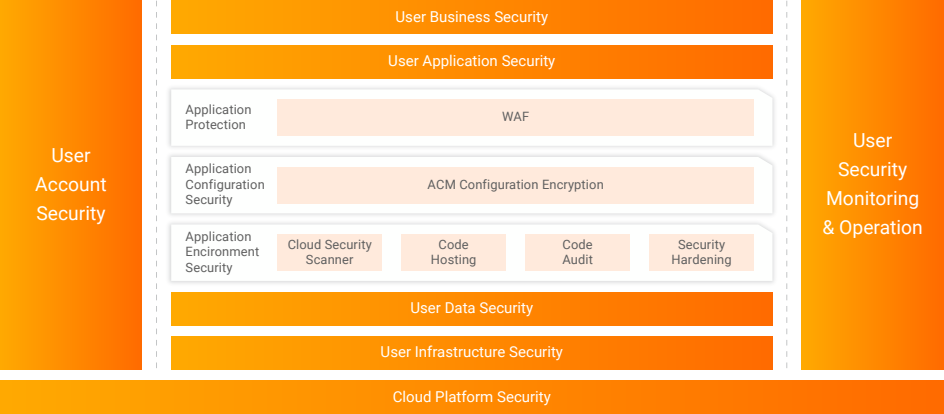


USER DATA SECURITY



User data security on the cloud is the lifeline of users and one of the most important representational attributes of the overall cloud security capability. In July 2015, Alibaba Cloud initiated the first Data Protection Proposal among cloud service providers in China. This public proposal appeals that the ownership of data of all developers, companies, governments, and social institutions on the cloud computing platforms belongs only to the users. Cloud computing platforms cannot use the data for other purposes. Cloud service providers have the responsibilities and obligation to help users ensure the confidentiality, integrity, and availability of their data.

USER APPLICATION SECURITY

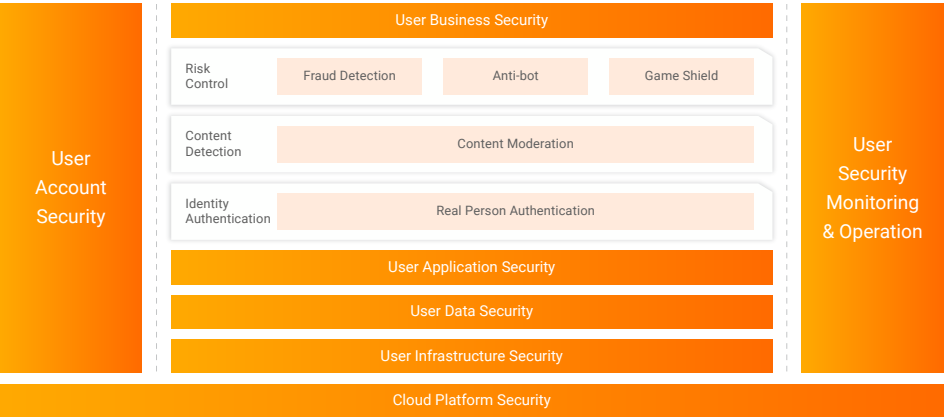


Applications built on Alibaba Cloud also need to be properly secured. At the application security level, Alibaba Cloud provides users with security capabilities in three aspects: application environment security, application configuration security, and application protection.



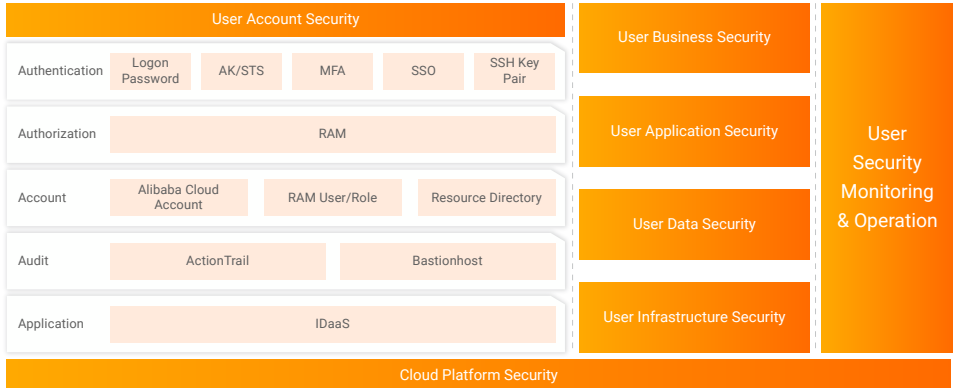


USER BUSINESS SECURITY



After a user's application has been built and is running on the Alibaba Cloud platform, Alibaba Cloud provides corresponding security capabilities for different business scenarios of the user. At the business security level, Alibaba Cloud provides security features from three aspects: identity authentication, content detection, and risk control.

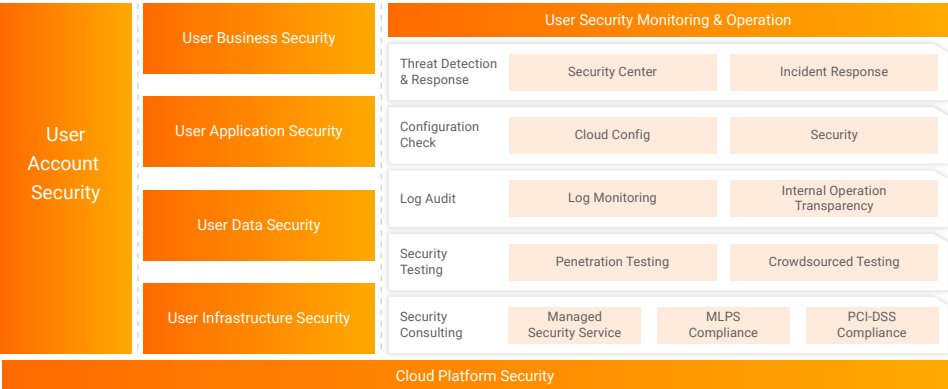
USER ACCOUNT SECURITY



User account security is an important dimension in the overall cloud security architecture design. User account security on the cloud involves the “Five A’s”: Authentication, Authorization, Account management, Audit, and Application management. It is important to note that in cloud-based business, there is often a cloud product or service (such as RAM) that provides multiple dimensions of capabilities in the aforementioned aspects. This chapter describes Alibaba Cloud account security capabilities specific to each dimension.



USER SECURITY MONITORING AND OPERATIONS



After a user's resources, applications, business, and accounts are properly protected on the cloud, the cloud platform needs to provide security monitoring and operations capabilities to detect security threats, monitor the configurations of cloud resources and logs, and, where appropriate, perform security testing on the user's overall business environment, as well as provide a response along with consulting services. In this way, the security monitoring and operations capabilities of Alibaba Cloud can effectively safeguard the user's business.

4.6.2 ANTI-DDOS

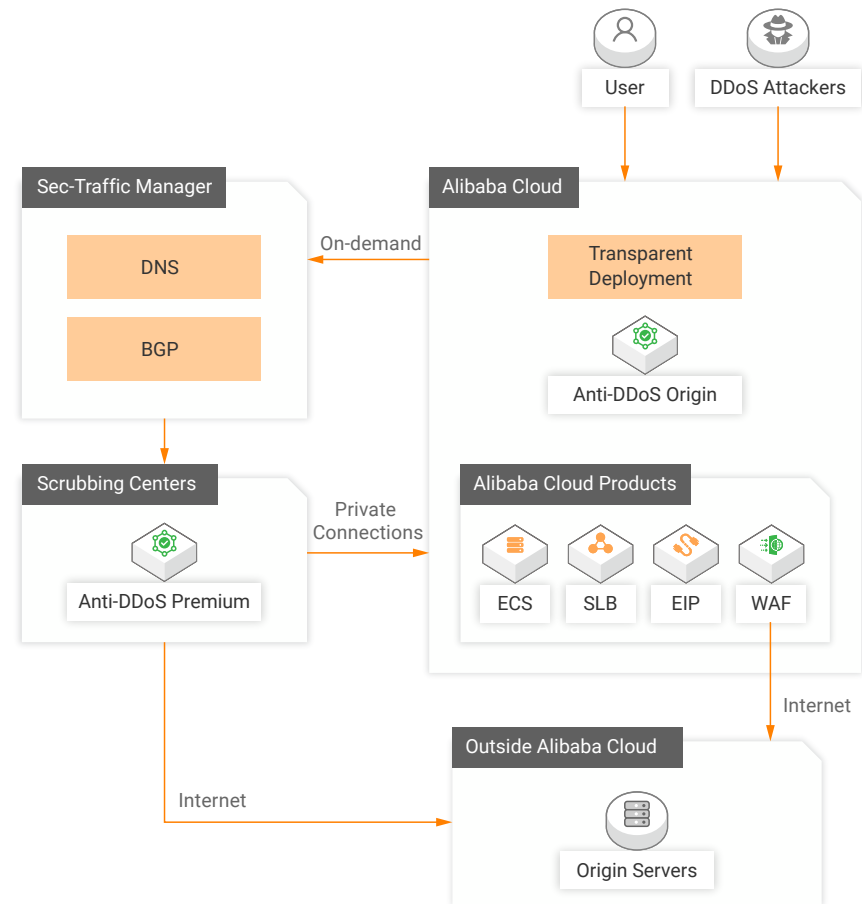
Anti-DDoS is provided for enterprises whose services on the Internet become unavailable due to DDoS attacks.

By default, Anti-DDoS provides an Anti-DDoS capacity of up to 5 Gb/s for free. In addition, Alibaba Cloud has launched the Security Credibility plan. After becoming a member of Security Credibility, you can enjoy the additional DDoS mitigation capacity on top of the default offering based on your security credibility score.



DDOS PROTECTION PACKAGE

The Anti-DDoS protection package is designed for large-scale enterprises on Alibaba Cloud. It uses the native Alibaba Cloud network and transparent protection engine to mitigate DDoS attacks. The protection package can be used to protect various Alibaba Cloud products, such as ECS, SLB, WAF, and EIP.



Procedure

1. Purchase an Anti-DDoS Origin Enterprise instance.

Anti-DDoS Origin Enterprise instances are only available in Mainland China. If you want to purchase an Anti-DDoS Origin Enterprise instance outside of Mainland China, you must submit a ticket or contact sales personnel. After the request is approved, you can purchase an Anti-DDoS Origin Enterprise instance.

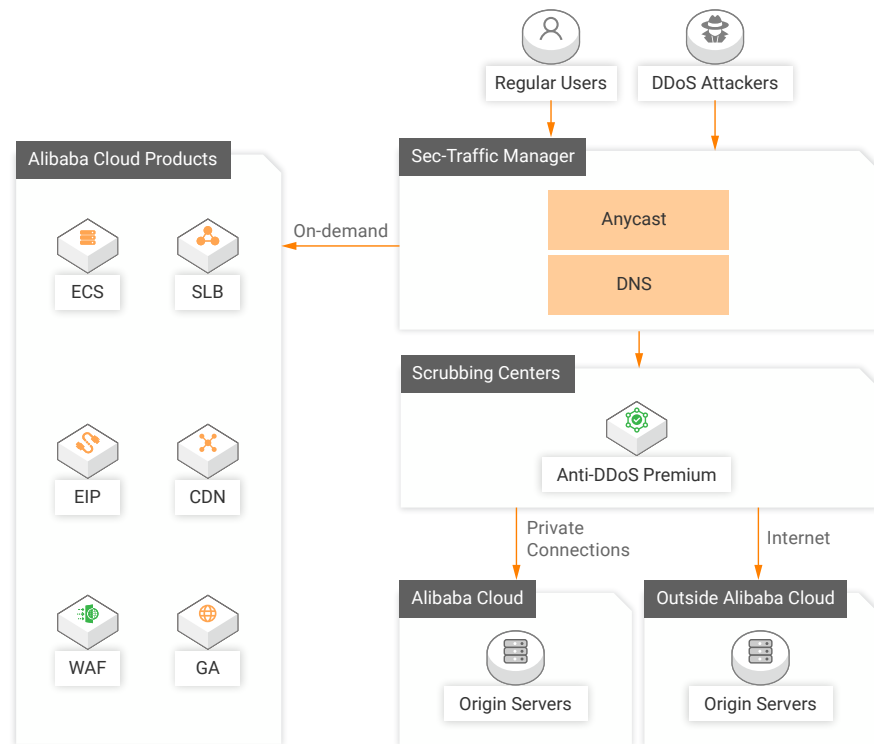
2. Log on to the Anti-DDoS Basic console.

3. On the Anti-DDoS Origin page, find the target instance and click 'Add a protection target' to add the addresses that you want to protect to the instance.



ANTI-DDOS PRO AND PREMIUM

Anti-DDoS Pro and Anti-DDoS Premium are designed for enterprises that deploy their resources on or off Alibaba Cloud. By using the massive traffic scrubbing center resources of Alibaba Cloud, both products work together with the artificial intelligence (AI) protection engine and adopt full-traffic proxy to protect against high-volume traffic attacks and refined web application level resource exhaustion attacks, such as Hypertext Transfer Protocol (HTTP) flood attacks.



Set up Anti-DDoS Pro using domains

1. Add forwarding rules

In the Anti-DDoS Pro or Anti-DDoS Premium console, add a website service that you want to protect by using a domain name, associate the service with an Anti-DDoS Pro or Anti-DDoS Premium instance, and configure the traffic forwarding rules.

2. Configure service traffic forwarding

Modify the DNS records of your domain name to reroute the traffic directed to your website to an Anti-DDoS Pro or Anti-DDoS Premium instance. The instance scrubs the traffic and then forwards the traffic to the origin server, which protects your website service against DDoS attacks.



3. Configure protection policies

After you set up an Anti-DDoS Pro or Anti-DDoS Premium instance to protect your website service, Intelligent Protection is enabled automatically. You can manually adjust Anti-DDoS protection policies for your website service, which includes Intelligent Protection, Black Lists and White Lists (Domain Names), Blocked Regions (Domain Names), Accurate Access Control, and Frequency Control.

4. View the protection data of your website services

After you set up an Anti-DDoS Pro or Anti-DDoS Premium instance to protect your website service, you can use the Security Reports feature and log-related features to view the protection data in the Anti-DDoS Pro or Anti-DDoS Premium console.

Set up Anti-DDoS Pro using IPs and ports

1. Create a port forwarding rule

Add a non-website service that you want to protect by using a port in the Anti-DDoS Pro or Anti-DDoS Premium console. Use the IP address of your Anti-DDoS Pro or Anti-DDoS Premium instance as your service IP address to reroute inbound traffic to your instance. After you change the IP address, the instance scrubs the inbound traffic and then forwards the traffic to the origin server.

2. Configure port forwarding and Anti-DDoS protection policies

Configure port forwarding policies as required, such as session persistence and health checks for multiple origin IP addresses. You can also configure Anti-DDoS protection policies for non-website services, such as False Source, Speed Limit for Destination, Packet Length Limit, and Speed Limit for Source.

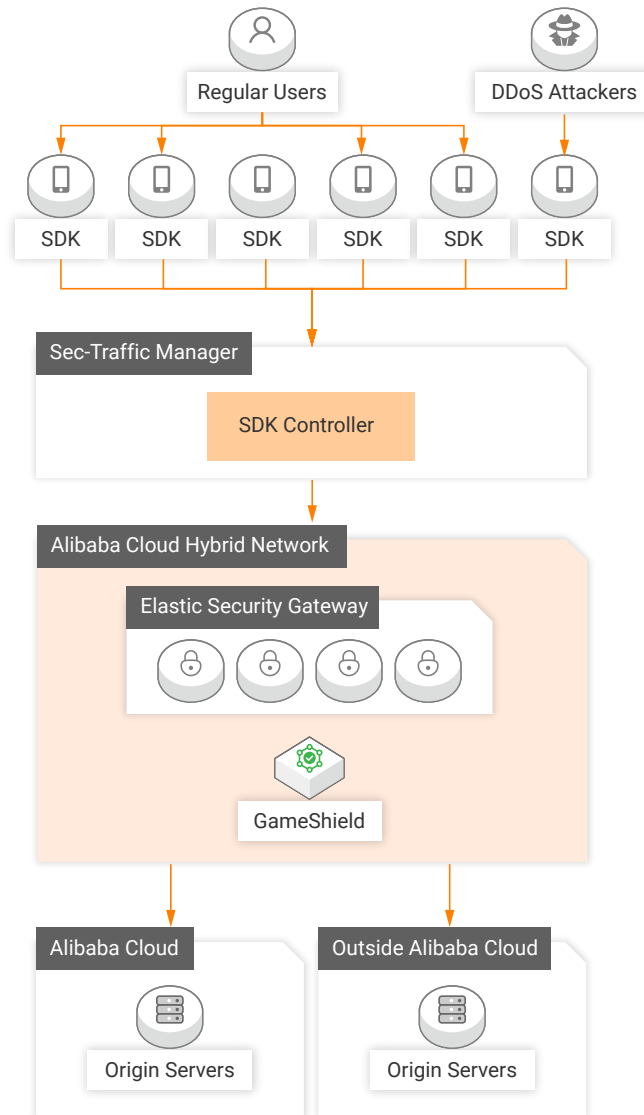
3. View the protection data of a port

View the traffic that goes through a port on the Security Overview page of the Anti-DDoS Pro or Anti-DDoS Premium console after you set up an Anti-DDoS Pro or Anti-DDoS Premium instance to protect your non-website service.



GAMESHIELD

GameShield is a network security solution presented by Alibaba Cloud for protecting the gaming industry against DDoS attacks. Compared with Anti-DDoS Pro, GameShield can defend against high-traffic DDoS attacks at the Tbps (TeraBits Per Second) level and TCP resource exhaustion attacks (Layer-4 CC – Challenge Collapsar - attacks) that are especially applicable to the gaming industry. This solution reduces protection costs and provides better protection.



Procedure

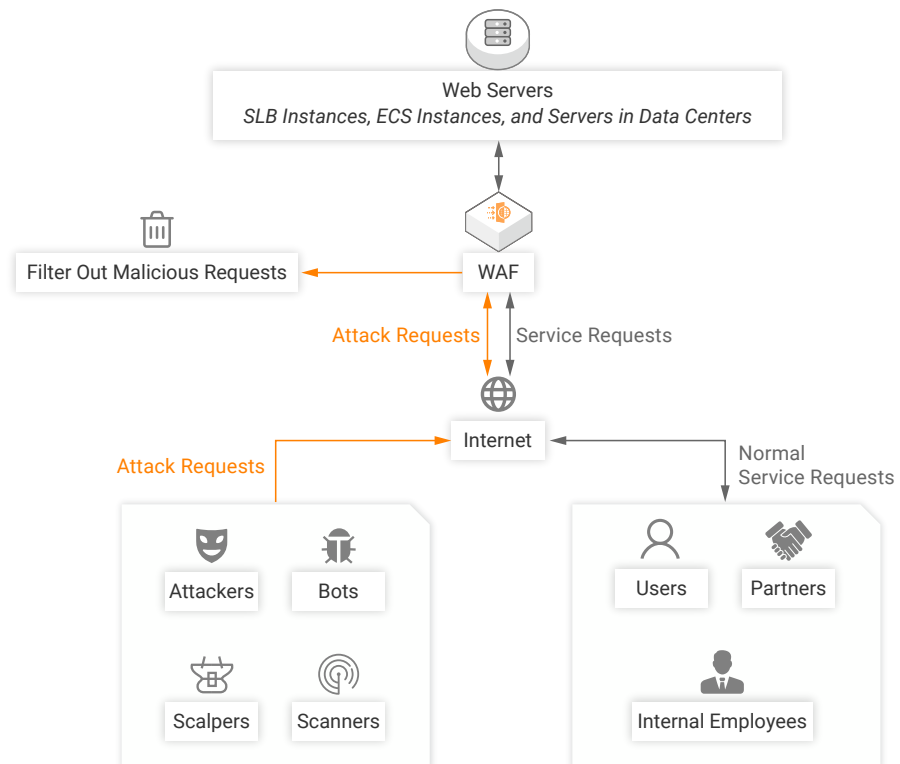
1. Create a game instance.
2. Configure a protection object.
3. Add your application to GameShield.



4.6.3 WAF

Based on the big data and intelligent computing capabilities of Alibaba Cloud Security, Alibaba Cloud WAF defends against common security threats reported by OWASP (Open Web Application Security Project). These threats include SQL injection, cross-site scripting (XSS), common vulnerabilities in web server plugins, web shell upload, and unauthorized access to cloud resources. WAF also filters out massive numbers of malicious access attempts, prevents leakage of web assets and data, and ensures the security and availability of web applications.

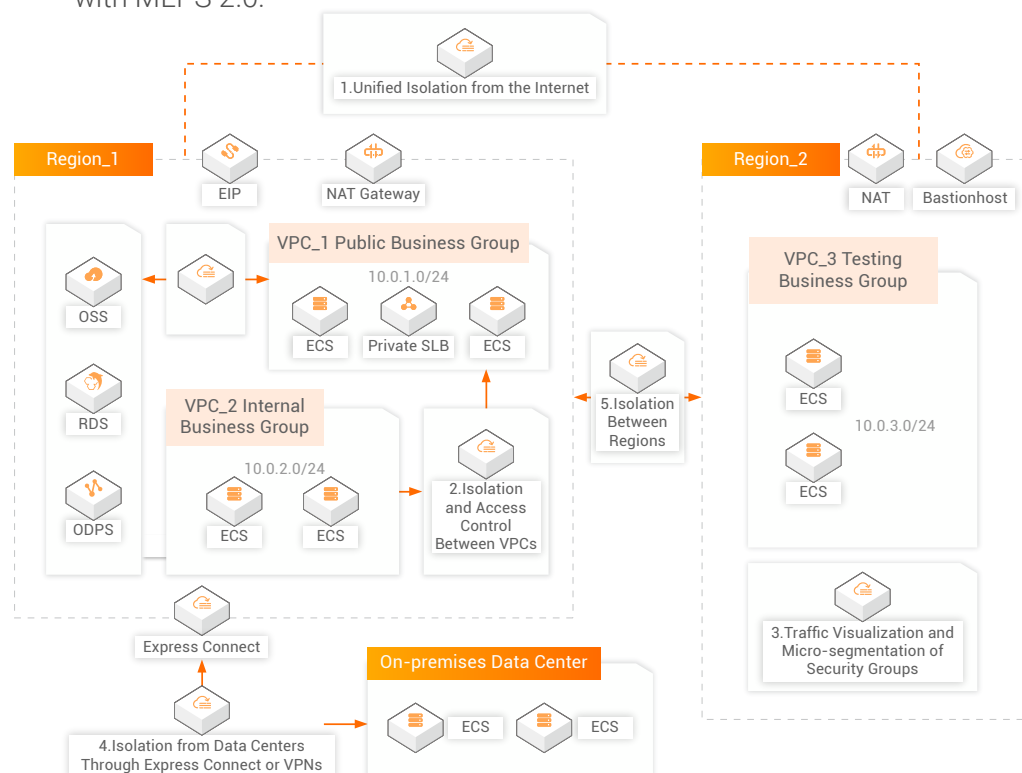
When you add your website to WAF in Canonical Name (CNAME) record mode, you must enter the website information and change the DNS record to resolve the domain name of your website to the CNAME assigned by WAF. This way, the requests destined for your website are redirected to WAF. This mode is supported regardless of whether your origin server is deployed on the cloud.



Alibaba Cloud offers Hybrid Cloud WAF to help protect and manage web applications in various hybrid cloud environments in a centralized manner. The web applications can be deployed across Alibaba Cloud Public Cloud and data centers, as well as across Alibaba Cloud Public Cloud and third-party clouds. In this case, Hybrid Cloud WAF protects your service traffic that cannot be migrated to the cloud. To deliver an elastic and effective system, Hybrid Cloud WAF combines shared and exclusive resources in and outside of the cloud. This way, you can protect the web applications in a centralized manner.

4.6.4 CLOUD FIREWALL

Cloud Firewall is the industry's first firewall as a service (FWaaS) solution for public cloud environments. This product manages the access control policy for north-south traffic from the Internet to ECS instances, and the micro-isolation policy for east-west traffic between ECS instances in a centralized manner. Cloud Firewall comprehensively checks the exposure of your assets to the Internet in the cloud environment, whilst managing the access policies for public network IP addresses in a centralized manner. With one-click access, Cloud Firewall is the primary network security infrastructure for ensuring the network security of your businesses that have been migrated to Alibaba Cloud. With the built-in intrusion prevention system (IPS), this product provides intelligent prevention, detects victim hosts, blocks external connections launched by hosts, and visualizes network-wide traffic and access relationships among ECS instances. Cloud Firewall meets the requirements of Multi-Level Protection Scheme (MLPS) 2.0 for virtual boundaries, access control, IPS, and storage time of security logs, which must be retained for at least six months. This product is a necessary choice to be compliant with MLPS 2.0.

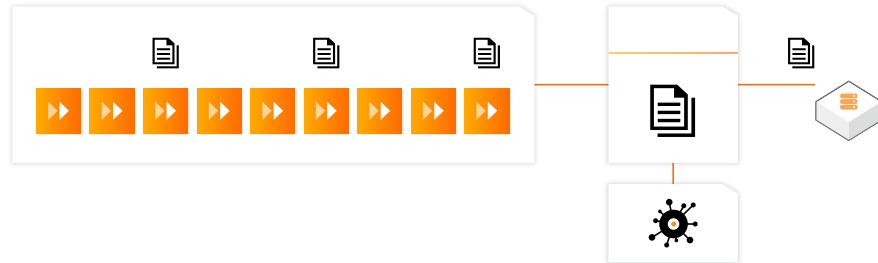


Procedure

1. On the Firewall Settings page, enable firewalls.
2. Choose Intrusion Prevention > Intrusion Prevention to configure intrusion prevention policies.
3. On the Access Control page, configure access control policies.
4. On the Traffic Analysis page, view traffic statistics. You can also view information about outbound connections.

4.6.5 SECURITY CENTER

Security Center is a unified security management system that identifies, analyzes, and alerts security threats in real-time. With security capabilities such as anti-ransomware, anti-virus, tamper-proofing, and compliance assessment, users can automate security operations, responses, and threat tracing to safeguard both cloud and local servers and meet regulatory compliance requirements.

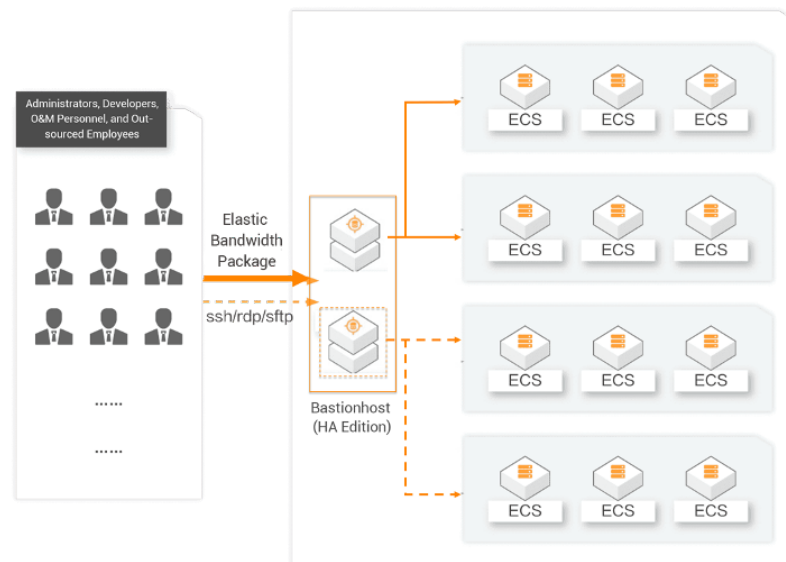


Procedure

1. Log on to the Security Center console.
2. In the Unprotected assets (such as ECS) section on the Overview tab, you can view the number of unprotected servers that do not have the Security Center agent installed.
3. Click Install to go to the Settings page. Click the Agent tab and then the 'Client to be installed' tab. On the tab that appears, you can view the total number and list of servers that do not have the Security Center agent installed.
4. Click the 'Client Installation Guide' tab to choose automatic installation or download the installation package and run commands to install the Security Center agent.

4.6.6 BASTION HOST

Bastion Host provides multiple functions, such as O&M identity authentication, account management, and system operation audit. Bastion Host acts as a forward proxy and records the data flow of common O&M protocols, such as Secure Shell (SSH), Remote Desktop Protocol (RDP), and Secure File Transfer Protocol (SFTP). It can also regroup the data flows and reproduce them to implement an O&M audit.



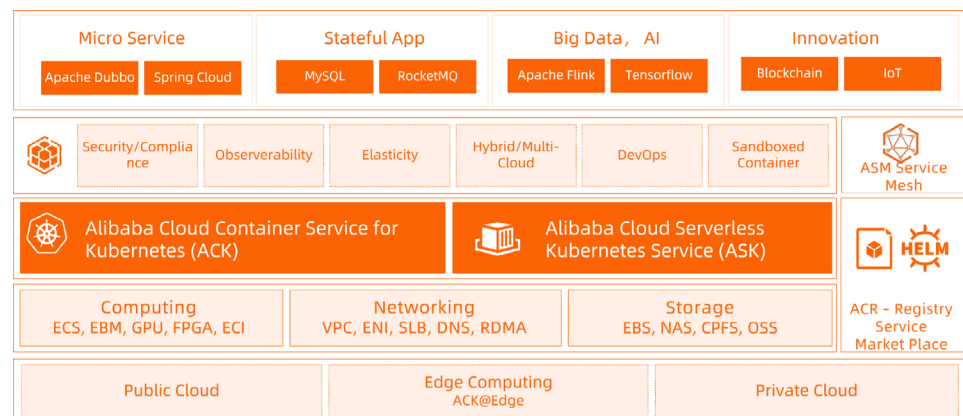
Procedure

1. Synchronize ECS assets.
2. Omport Alibaba Cloud RAM users.
3. Create O&M rules.
4. Perform O&M operations on hosts.
5. Audit O&M sessions.

4.7 CONTAINER

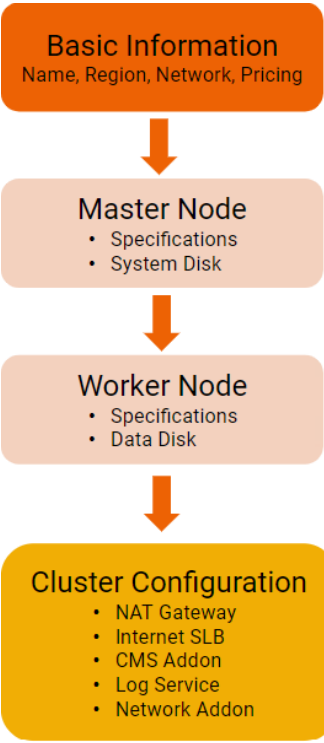
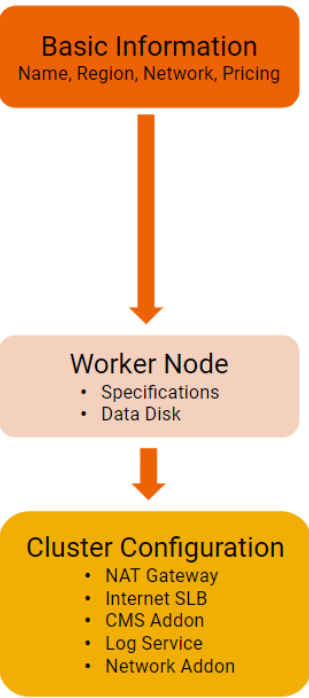
Alibaba Cloud Container Service for Kubernetes (ACK) is one of the first services to pass the Certified Kubernetes Conformance Program. ACK provides high-performance management services for containerized applications. You can manage enterprise-level containerized applications throughout the application lifecycle.

The following figure shows the architecture of the Alibaba Cloud Container Service product portfolio.



ACK provides the following three cluster types:

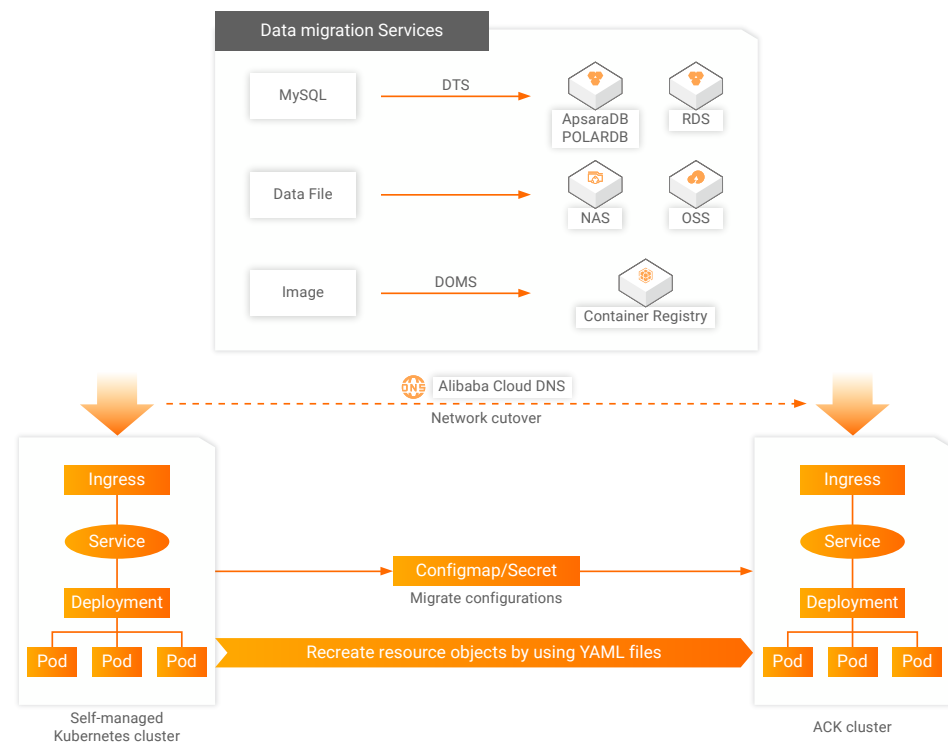
- » Dedicated Kubernetes clusters,
- » Managed Kubernetes clusters
- » Serverless Kubernetes clusters.

Item	Dedicated Kubernetes cluster	Managed Kubernetes cluster	Serverless Kubernetes
Feature	You must create master nodes and worker nodes. A dedicated Kubernetes cluster allows you to manage the cluster infrastructure in a more fine-grained manner. You must design, maintain, and upgrade the Kubernetes cluster on your own.	You only need to create worker nodes. ACK creates and manages master nodes. A managed Kubernetes cluster is easy-to-use, cost-efficient, and highly available. You do not need to manage master nodes.	A serverless Kubernetes cluster allows you to start applications directly. You do not need to manage nodes.
Billing method	You are not billed for cluster management features. However, you must pay for master nodes, worker nodes, and other infrastructure.	Standard ACK clusters: You are not billed for cluster management features. However, you must pay for worker nodes and other infrastructure. Professional ACK clusters: You are billed based on the number of clusters or the subscription method.	You are billed for the resources consumed by pods and the resource usage duration. The duration is measured in seconds.
Use scenario	Applies to all scenarios.	Applies to all scenarios.	Applies to batch tasks, urgent application scale-out, and continuous integration (CI) or continuous delivery (CD).
User profile	» Cost insensitive » Familiar with Kubernetes » Maintenance capable » Clear resource planning » Self management » Master node customized	» Cost reduction » Focus on application » New to Kubernetes » Few Kubernetes maintenance » No care on master node	» In-batch task » Excellent scalability » Run out of the box, and stop immediately » Zero maintenance » Pay as you go » No care on IaaS
Cluster creation procedure	 <pre> graph TD A[Basic Information Name, Region, Network, Pricing] --> B[Master Node • Specifications • System Disk] B --> C[Worker Node • Specifications • Data Disk] C --> D[Cluster Configuration • NAT Gateway • Internet SLB • CMS Addon • Log Service • Network Addon] </pre>	 <pre> graph TD A[Basic Information Name, Region, Network, Pricing] --> B[Worker Node • Specifications • Data Disk] B --> C[Cluster Configuration • NAT Gateway • Internet SLB • CMS Addon • Log Service • Network Addon] </pre>	 <pre> graph TD A[Basic Information Name, Region, Network, Pricing] --> B[Create Application] </pre>

HOW TO MIGRATE A GCP GKE MANAGED KUBERNETES CLUSTER TO ACK WITHOUT SERVICE DISRUPTION

As GCP GKE provides a standard Kubernetes cluster, we can treat it as a self-managed Kubernetes.

MIGRATION SCHEME



Procedure

1. Create and configure an ACK cluster

O&M engineers create an ACK cluster and configure cluster resources. This step makes it easy for developers to migrate applications to Kubernetes.

For more information, see [Create a dedicated Kubernetes cluster](#).

The following parameters are required when you create the ACK cluster:

- » **Cluster type:** Select one of the following cluster types when you create the ACK cluster.
 - *Standard dedicated Kubernetes cluster* - You need to create and manage the master and worker nodes of the ACK cluster. You have full control over the ACK cluster and are charged for the resources used by the master and worker nodes.

- *Standard managed Kubernetes cluster* - You need to create and manage only the worker nodes of the ACK cluster. ACK creates and manages the master nodes and you are charged for only the resources that are used by the worker nodes.
- » **Operating system:** Select an operating system based on your requirements.
 - By default, the CentOS 7.6 or Aliyun Linux 2.1903 operating system is selected. Alibaba Cloud recommends that you keep the default settings.
 - If the default system kernel cannot meet your requirements, you can use a custom image. For more information, see [Use a custom image to create an ACK cluster](#).
- » **VPC:** Select a virtual private cloud and vSwitches for the ACK cluster.
- » **SNAT:** Configure SNAT entries for the VPC.
- » **Public access:** Expose the API server of the ACK cluster with an EIP.
- » **Cloud Monitor agent:** Install the Cloud Monitor agent on the ECS instances of the ACK cluster.
- » **Log Service:** Install and configure the Log Service agent in the ACK cluster.

2. Migrate data

- » **Migrate databases.** For more information, see [Migrate data from a user-created MySQL database to an ApsaraDB RDS for MySQL instance](#).
 - a. Create a RDS instance.
 - b. Configure a RDS whitelist to allow only specified IP addresses to access the RDS instance.
 - c. Configure a PrivateZone. Use the PrivateZone to resolve the domain name of the database to the IP address of the RDS instance. This saves you the need to modify the database configuration on applications.
 - d. Migrate the data in the MySQL database of the self-managed Kubernetes cluster. Use Data Transmission Service (DTS) to migrate the data from the MySQL database to the ApsaraDB for RDS database in full, incremental, or two-way synchronization mode.



» **Migrate data.** For more information, see [Architectures and configurations](#).

- a. Activate OSS.
- b. Create an OSS bucket.
- c. Migrate the data that is stored in the self-managed Kubernetes cluster. Use the ossimport tool to migrate the data in batches from an on-premises server or a third-party cloud storage service to OSS. For example, you can choose a third-party cloud storage service such as Amazon S3, Microsoft Azure, or Tencent Cloud Object Storage.

» **Migrate images.** For more information, see [Use image-syncer to migrate container images](#).

- a. Create a Container Registry repository.
- b. Set the credential that is used to access the created Container Registry repository.
- c. Migrate the images of the self-managed Kubernetes cluster. You can use the image-syncer tool to migrate the images of the self-managed Kubernetes cluster to the Container Registry repository.

3. Migrate application configurations

ACK allows you to back up and restore stateful applications deployed in an ACK cluster. This is an all-in-one solution to achieve crash consistency, application consistency, and cross-region DR for stateful applications in ACK clusters.

For more information, see [Migrate applications to an ACK cluster](#).

Prerequisites: Two Kubernetes clusters of the same version are created.

For more information, see [Install the application backup component](#).



Method 1: Migrate an application by using the application backup feature

- a. Grant the required [OSS permissions](#) to `Cluster_A` and `Cluster_B`.
- b. [Install the application backup component](#) in `Cluster_A` and `Cluster_B`.
- c. [Create backup vaults](#) for `Cluster_A` and `Cluster_B`. When you create backup vaults for `Cluster_A` and `Cluster_B`, select the same OSS bucket and set Network Type to Public Network.
- d. [Create a backup task](#) in `Cluster_A`. After you create the backup task in `Cluster_A`, you can log on to `Cluster_B` to view the backup task on the **Backups** tab of the **Application Backup** page.
- e. On the **Application Backup** page, click the **Restore** tab and [create a restoration task](#) based on the backup task that is created in `Cluster_A`.

Method 2: Migrate an application by using Velero

This example shows how to migrate an application in the Nginx-examples namespace from `Cluster_A` to `Cluster_B`. To migrate the application, perform the following steps:

- a. Install the [Velero](#) client and the [application backup component](#) in `Cluster_A`. Then, configure and connect to the OSS bucket named `mybackups`.
- b. Install the Velero client and migrate-controller in `Cluster_B`. Then, configure and connect to the OSS bucket named `mybackups`.
- c. Back up applications in the **nginx-examples** namespace in `Cluster_A` and specify the name of the backup file, for example, **nginx-backup**.
- d. Use the backup file named **nginx-backup** in `Cluster_B` to restore the applications. This operation migrates the applications from `Cluster_A` to `Cluster_B`.
- e. **Perform regression tests.** Test engineers perform regression tests on the ACK cluster without interrupting the online business:
 - Configure the domain names for regression tests.
 - Test applications.
 - Check the logged application events.
 - Check the monitoring metrics of the applications.



- f. Switch all production traffic to ACK.** O&M engineers modify the DNS configuration to switch traffic to the ACK cluster:
 - Use the DNS service: Modify the DNS configuration to switch the traffic.
 - Upgrade the configurations or code of the client to switch the traffic.
- g. Bring the self-managed Kubernetes cluster offline.** O&M engineers check whether the ACK cluster can be accessed as expected. Then, O&M engineers bring the self-managed Kubernetes cluster offline:
 - Check whether the ACK cluster can receive and send traffic as expected.
 - Bring the self-managed Kubernetes cluster offline.
 - Clear the backup files stored in the OSS bucket that is created when you migrate applications.

4.8 MICRO SERVICES

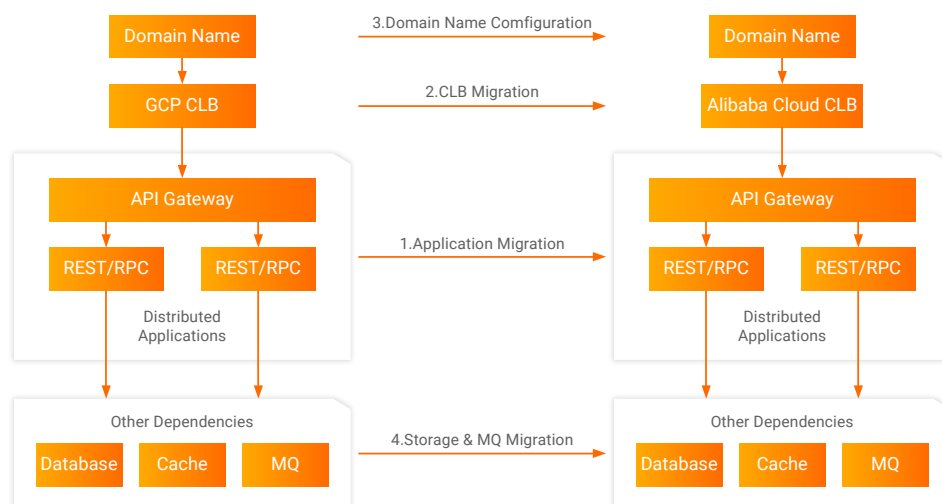
EDAS is an Application Platform as a Service (aPaaS) offering that allows you to host applications and manage microservices. EDAS provides full-stack solutions to develop, deploy, monitor, and maintain applications, and supports microservice frameworks such as Dubbo and Spring Cloud, which is a good choice when migrating microservices-based applications from GCP to Alibaba Cloud.

Since your applications have been deployed to the production environment and are running in a normal state, it is the first priority to ensure the smooth operation of the business without interruption during the migration process. At this point, Alibaba Cloud can provide very good support in this scenario, which can ensure that the application platform runs uninterrupted during the migration process.

4.8.1 MIGRATION PROCEDURE

Specifically, Alibaba Cloud supports two kinds of application migration of Spring Cloud and Dubbo microservice framework. The migration process of these two types of applications is the same, mainly including the following four aspects:





1. Application Migration

Applications to be migrated are typically stateless and can be migrated first. For more details, please refer to the following documents.

- » [Spring Cloud](#)
- » [Dubbo](#)

2. CLB Migration

If the GCP CLB instance is used before migration, then after the application migration is complete, you need to migrate from CLB to SLB instance on Alibaba Cloud.

If no GCP CLB instance is used before migration, it's recommended that you create and bind a SLB instance to the application after migration.

3. Domain Name Configuration

To create and bind a new SLB instance on Alibaba Cloud to migrated applications, you must add this SLB instance configuration to the domain name configuration. For more information, see [Change DNS servers for a domain name](#).

4. Storage and MQ Migration (Optional)

If your applications that deployed on GCP App Engine using other GCP services, such as Cloud SQL for MySQL and Pub/Sub Lite, you still need to migrate storage and message queues along with applications. For this part, you can follow [Chapter 2.4](#) database migration.



4.8.2 MIGRATION SOLUTION

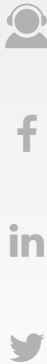
You can migrate applications by using two methods: registry switching, and dual registration and dual subscription. The two methods allow you to migrate applications during runtime without service discontinuity.

Microservice frameworks on GCP	Migration Solution	Description
Spring Cloud	Registry switching	Use Spring Cloud to switch the original service registry to Nacos. Develop applications and deploy them to EDAS, and switch traffic by configuring the SLB and the domain name. Refer to this document for more details.
	Dual registration and dual subscription	Access the original service registry and the EDAS service registry when applications are migrated to enable mutual calling between migrated applications and non-migrated applications. Refer to this document for more details.
Dubbo	Registry switching	Use Dubbo to switch the original service registry to EDAS Config Server and develop a new set of applications before deploying them to EDAS. Then, use SLB and configure domain names to switch traffic. Refer to this document for more details.
	Dual registration and dual subscription	Access the original service registry and the EDAS service registry when applications are migrated to enable mutual calling between migrated applications and non-migrated applications. Refer to this document for more details.

4.9 SERVERLESS

Alibaba Cloud Function Compute (FC) is a fully managed, event-driven compute service and is the best replacement for GCP Cloud Function.

Function Compute allows customers to migrate codes without any change from GCP Cloud Function. It also integrates the container ecosystem. It provides image deployment and supports a seamless switch from the original container deployment method.



FUNCTION COMPUTE CONTAINER IMAGE DEPLOYMENT

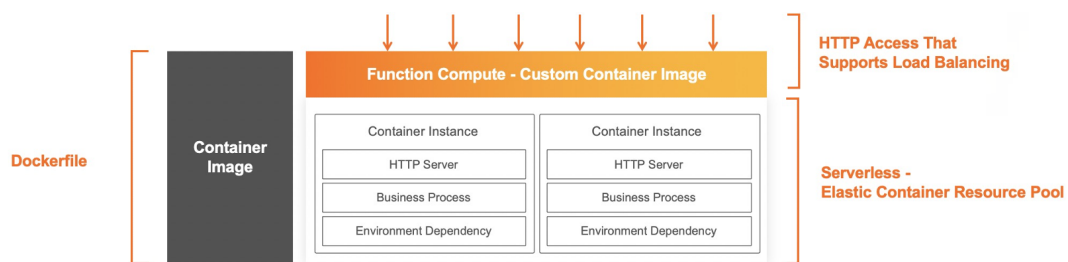
Function Compute integrates the container ecosystem. It provides image deployment and supports a seamless switch from the original container deployment method.

Disadvantages of conventional migration methods of standalone web applications (Spring Boot, WordPress, Flask, Express, and Rails):

- » O&M costs are increased to cater to modernized fine-grained responsibility division and service governance.
- » Historical Assets such as on-cloud or off-cloud code, dependencies, and configurations make it difficult to go serverless.
- » You need to plan capacity and build O&M and monitoring systems.
- » Exclusive resources are allocated to low-traffic services, which causes low resource usage.

Advantages of application migration based on Function Compute and container images:

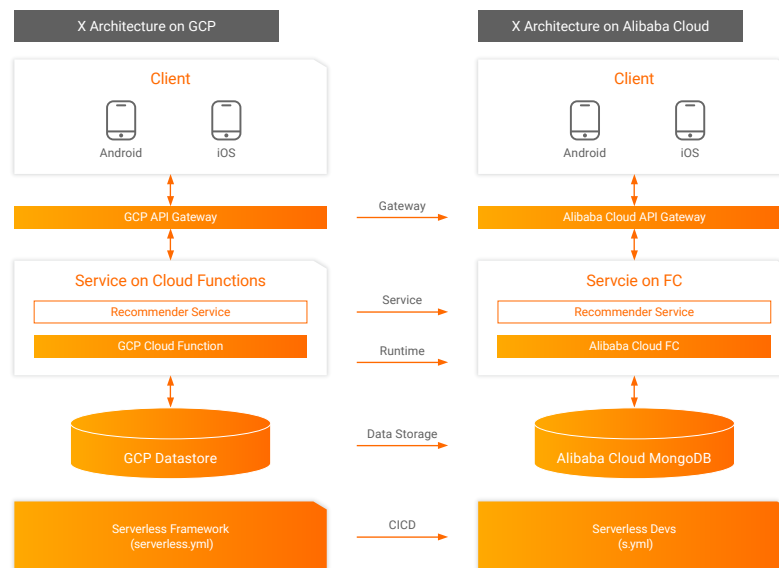
- » This method migrates standalone applications at low costs.
- » This method does not require O&M.
- » This method supports auto scaling. You do not need to arrange capacity.
- » This method ensures that all resources are in use, achieving 100% resource usage and reducing IT costs.



Serverless integrates the capabilities of the cloud ecosystem based on SDKs and APIs. FC and Cloud Function both integrate into the cloud ecosystem based on the unique event-driven callbacks.

As shown below, Alibaba Cloud has the most complete product matrix compared with GCP, so if the customer's services depend on other cloud products, the migration will also be easy.

	 Alibaba FC	 GCP Cloud Function	 Tencent SCF
Storage	• OSS • Log Service	• Cloud Storage • Cloud Logging	• COS • CLS
NoSQL DB	• TableStore	• Datastore • Firestore	- Not supported
Messaging	• Kafka • RocketMQ • MNS queue • MNS Topic	• Firebase	• Kafka
IoT	• IoT Platform • IoT Link Edge • MQTT	• Cloud IoT	- Not supported
Event Hub	• EventBridge	• Pub/Sub	- Not supported
CDN Origin	• Built-in support	• Via API Gateway	• Via API Gateway
Monitoring & Operations	• Cloud Monitor • OOS	• Cloud Logging • VM Manager	- Not supported
Orchestration	• Serverless Workflow • ROS	• Cloud Composer • Cloud Deployment Manager	• Application Service Workflow (in Beta, no international version)
Outbound Integrations	• EventBridge • FC • MNS Queue • MNS Topic	• - Not supported	- Not supported



4.10 MESSAGE QUEUES

4.10.1 MESSAGE QUEUE FOR MQTT

AliwareMQ for IoT is a messaging service that is designed specifically for IoT and mobile Internet (MI). The service can be applied to multiple scenarios, including live streaming, financial payments, smart catering, instant messaging, mobile apps, smart devices, and the Internet of Vehicles. Message Queue for MQTT (Message Queue Telemetry Transport) supports multiple protocols including MQTT and WebSocket to enable two-way communications between devices and the cloud. In this way, messages can be exchanged in multiple business models including Consumer-to-Consumer (C2C), Consumer-to-Business (C2B), and Business-to-Consumer (B2C) to achieve the Internet of Everything.

LIMITATIONS:

AliwareMQ for IoT is not responsible for migrating:

- » Metadata (ex. topic)
- » Server Data

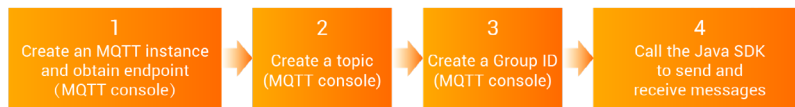
This means that when using AliwareMQ for IoT, the Topic data needs to be recreated on AliwareMQ for IoT.

1. Comparison between Message Queue for IoT and GCP IoT Core:

Category	Item	AliwareMQ for IoT	GCP IoT Core
Protocols	Accessing protocols	MQTT3.1.1, WebSocket, TLS, HTTP	MQTT, HTTP, TLS
Message	Message model	pub/sub, p2p	pub/sub
Security	Devices certification	Signature certification, unique-certificate-per-device, Token	JWT, Token
	Rights management	Different accounts have different metadata management and data access permissions	Supported
	Devices management	Unsupported	Supported
Logging and monitoring	Monitoring resources	Supported	Supported
Important Quotas and limits	Device-to-cloud throughput, per device	Unlimited	512KB/s
	Cloud-to-device throughput, per device	Unlimited	512KB/s
	Devices attached to a gateway	Unlimited	10000

2. Choose Migrate GCP IoT Core to AliwareMQ for IoT

The following figure shows the procedure of using AliwareMQ for IoT SDK for Java to send and subscribe to messages:



1. Create an AliwareMQ for IoT instance and obtain its endpoint.
2. Create a parent topic. In this step, you can create a topic of the same name with GCP IoT Core.
3. Create a group ID. This step is unique to AliwareMQ for IoT. Groupid is a flag used to distinguish a group of devices.
4. Send and subscribe to messages by using AliwareMQ for IoT SDK for Java.

Enable devices using MQTT protocol on GCP IoT Core to connect to the AliwareMQ for IoT server and pub or sub messages. You need to modify the SDK configuration parameters:

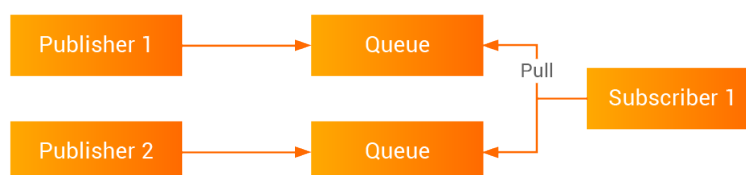
- » Configuring instancelid.
- » Update endpoint.
- » Configuring accessKey and secretKey.
- » Configuring clientId as AliwareMQ for IoT define

4.10.2 MESSAGE SERVICE

Alibaba Cloud MNS (Managed Network Services) is a distributed messaging service that provides scalability, high efficiency, reliability, security, and availability. MNS allows developers to transfer data and notifications between the distributed components of an application and build a loosely coupled system. MNS provides a queue-based messaging model and a topic-based messaging model.

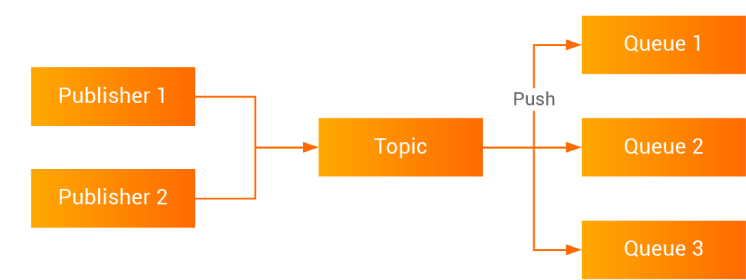
Message Service provides a more convenient queue mode and message communication can be completed without Topic.

Queue Mode



At the same time, the Topic mode is provided to distribute messages.

Topic Mode



COMPARISON BETWEEN MESSAGE SERVICE AND GCP IOT CORE

Category	Item	Message Service	GCP Pub/Sub
Message	Message Model	Topic, Queue	Topic
	Batch	Supported	Unsupported
	Long Polling Period	Supported	Unsupported
	Visibility Timeout Period	Supported	Unsupported
	Scheduled Period	Supported	Unsupported
Important Quotas and limits	Publisher throughput	Unlimited	120,000,000 KB per minute (2 GB/s) in large regions
	Pull subscriber throughput	Unlimited	240,000,000 KB per minute (4 GB/s) in large regions
	Message size	256 KB	Message size (the data field): 10MB

4.11 DEVOPS AND AUTOMATION (TERRAFORM)

Terraform is an infrastructure as code (IaC) tool that allows you to build, change, and version infrastructure safely and efficiently. This includes low-level components such as compute instances, storage, and networking, as well as high-level components such as DNS entries, SaaS features, etc. Terraform can manage both existing service providers and custom in-house solutions.

The official Alibaba Cloud team is responsible for the daily development and maintenance of the Terraform Provider. If you have used AWS/Azure/GCP providers before, there will not be much change in the main framework, syntax, and usage habits. The main difference is the definition of products, resources, and business use. You can obtain detailed information about these contents through [Alibaba Cloud official documents](#).

The following table prepares the correspondence between commonly used products and resources for you so that you can quickly compare and find the corresponding document links.

Alicloud Product Name	TF Product/TF Resource Sample	GCP Product Name	TF Product / TF Resource Sample
Container Service for Kubernetes (ACK)	Container Service for Kubernetes (ACK) / <code>alicloud_cs_autoscaling_config</code>	Google Kubernetes Engine (GKE)	GKEHub / <code>google_gke_hub_feature</code>
Container Registry (ACR)	Container Registry (CR) / <code>alicloud_cr_ee_instance</code>	Container Registry	Container Registry / <code>google_container_registry</code>
Function Compute (FC)	Function Compute Service / <code>alicloud_fc_alias</code>	Cloud Functions	Cloud Functions / <code>google_cloudfunctions_function</code>
AnalyticDB for MySQL	RDS / <code>alicloud_db_instance</code>	BigQuery	BigQuery / <code>google_bigquery_dataset</code>
ApsaraDB for Redis	Redis And Memcache (KVStore) / <code>alicloud_kvstore_instance</code>	Cloud MemoryStore	Memorystore (Redis) / <code>google_redis_instance</code>
ApsaraDB RDS for MySQL	RDS / <code>alicloud_db_instance</code>	Cloud SQL	Cloud SQL / <code>google_sql_database</code>
Elastic Compute Service (ECS)	ECS / <code>alicloud_instance</code>	Compute Engine	Compute Engine / <code>google_compute_instance</code>
Auto Scaling	Auto Scaling (ESS) / <code>alicloud_ess_alarm</code>	Compute Engine Autoscaler	Compute Engine / <code>google_compute_region_autoscaler</code>
Resource Orchestration Service (ROS)	ROS / <code>alicloud_ros_template</code>	Cloud Deployment Manager	Cloud Deployment Manager / <code>google_deployment_manager_deployment</code>
Virtual Private Cloud (VPC)	VPC / <code>alicloud_network_acl_attachment</code>	Cloud Virtual Network	Compute Engine / <code>google_compute_shared_vpc_host_project</code>
Server Load Balancer (SLB)	Classic Load Balancer (CLB) / <code>alicloud_slb</code>	Cloud Load Balancing	AssuredWorkloads / <code>google_assured_workloads_workload</code>
VPN Gateway	VPN / <code>alicloud_vpn_customer_gateway</code>	Cloud VPN	Compute Engine / <code>google_compute_external_vpn_gateway</code>
Anti-DDoS	Anti-DDoS Pro / <code>alicloud_ddosbgp_instance</code>	Cloud Security Scanner	Cloud Security Scanner / <code>google_security_scanner_scan_config</code>
Web Application Firewall (WAF)	Web Application Firewall (WAF) / <code>alicloud_waf_instance</code>	Cloud Armor, reCAPTCHA Enterprise Apigee	Apigee / <code>google_apigee_envgroup</code>
Resource Access Management (RAM)	RAM / <code>alicloud_ram_access_key</code>	Identity and Access Management	Cloud IAM / <code>google_iam_workload_identity_pool</code>

Alicloud Product Name	TF Product/TF Resource Sample	GCP Product Name	TF Product / TF Resource Sample
Elastic Block Storage (EBS)	ECS / <code>alicloud_disk</code>	Compute Engine	Compute Engine / <code>google_compute_disk</code>
Object Storage Service (OSS)	OSS / <code>alicloud_oss_bucket</code>	Cloud Storage	Cloud Storage / <code>google_storage_bucket</code>
File Storage NAS	Network Attached Storage (NAS) / <code>alicloud_nas_access_group</code>	File Store	Filestore / <code>google_filestore_instance</code>
Log Service	Log Service (SLS) / <code>alicloud_log_project</code>	Pub/Sub	Cloud Pub/Sub / <code>google_pubsub_lite_reservation</code>

For more information about Terraform hands-on, please refer to [this webpage](#).

ABOUT

Established in 2009, Alibaba Cloud (alibabacloud.com), the digital technology and intelligence backbone of Alibaba Group, is among the world's top three IaaS providers, according to Gartner. It is also the largest provider of public cloud services in China, according to IDC.

Alibaba Cloud provides a comprehensive suite of cloud computing services to businesses worldwide, including merchants doing business on Alibaba Group marketplaces, start-ups, corporations and public services.

Alibaba Cloud is the official Cloud Services Partner of the International Olympic Committee.

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